3571 Cassie Drive SW

기존 3-car garage → AFH 거주구역 전환 (침실 3 + 통합욕실)

1. 이 세트는 Canarch Designs A01.3(기존)/A01.4(제안) 기반 스키매틱 BIM/도면입니다.
2. 실 치수는 A01.4 표기값을 사용했고, 벽 위치는 그 치수에 맞춘 배치입니다.
3. 제출 전 추가 필요: 대지/주차 계획, 구조 검토, WSEC 계산, M/E/P 실시도,
연기·CO 경보 배치도, AFH(DSHS WAC 388-76) 검토, Owner-builder 선언서.
4. 천장고 10'-5". 기존 차고도어 유지 + 내부 스킨월 마감(A01.4 주기).
5. 창호: 신규 5x4 슬라이더 3개(창턱 40"), 침실별 이그레스 확보 목적.
6. 자동검사 결과는 report.html 참조. 검사 통과는 법규·구조 안전을 보증하지 않습니다.

3571 Cassie Drive SW | Tumwater, WA 98512

LOT 7, CHAPPAREL LANDING DIVISION 2 · Tax-ID 4038-00-00700

AFH 전환 (기존 3-car garage) - 허가 제출용 스키매틱

원안: Canarch Designs (Caleb Mongare) - A01.3 / A01.4

Project No. CMN/25/122 Date OCT 2026

SHEET G-00 표지 및 일반주기

이 도면은 A01.3/A01.4 기반 스키매틱입니다.
시공 전 실측 확인 필요. owner-builder 제출, 서명 불요.

1. 적용 코드: 2021 IRC (WA 개정), 2021 WSEC, Tumwater 시 조례, WAC 388-76(AFH)

2. 이 세트는 Canarch Designs A01.3/A01.4 기반 스키매틱이며 시공용이 아님

3. 모든 치수는 공적기록과 현장 실측으로 확인 후 사용

4. 구조: 기존 외벽 유지, 신규 파티션 비내력. 헤더·개구부는 구조기술사 확인

5. 소방: 연기/CO 상호연동 + 배터리 백업 (WAC R314.4, IRC R314.3/R315.5)

6. 이그레스: 침실 3개소 이그레스 창 확보 (IRC R310, 순개구부 5.7 sqft)

7. 에너지: WSEC 2021 외피 성능(EN-01 참조), 기밀 테스트 별도

8. 설비: M/E/P 는 예비안. 최종 관경·용량은 해당 설계자 확정

9. AFH: 침실 면적·문폭·화장실 접근성은 DSHS 라이선스 기준 별도 검토

10. Own-builder 제출 (시 담당자 확인: 건축사·구조기술사 서명 불요)

11. 가스 온수기가 차고 내부에 설치 — 거주공간 전환 시 이전 또는 연소공기·격리 조치 필요 (IRC M1307 / G2408)

이 도면 세트는 예비 검토용입니다.

3571 Cassie Drive SW | Tumwater, WA 98512

LOT 7, CHAPPAREL LANDING DIVISION 2 · Tax-ID 4038-00-00700

AFH 전환 (기존 3-car garage) - 허가 제출용 스키매틱

원안: Canarch Designs (Caleb Mongare) - A01.3 / A01.4

Project No. CMN/25/122 Date OCT 2026

SHEET G-01 일반주기 및 코드 요약

이 도면은 A01.3/A01.4 기반 스키매틱입니다.
시공 전 실측 확인 필요. owner-builder 제출, 서명 불요.

SHEET	제목
G-00	표지
G-01	일반주기
G-02	범례/목록
A-01	기존 평면
A-02	제안 평면
A-05	치수 평면
A-06	마감 평면
A-07	반사천장
A-08	가구 배치
A-13	단면 A-A
A-14	단면 B-B
A-15..18	상세 4종
S-01	구조 계획
S-02	기초/슬래브
S-03	지붕/장선
M-01	설비 개념
M-02	기계 평면
M-03	덕트 계통
E-01	전기 평면
E-02	조명 평면
E-03	동력 결선
P-02	급배수 평면
P-03	급수 계통
P-04	배수/통기
LS-01	연기/CO
LS-02	이그레스
EN-01	외피 요약
EN-02	창호 성능

범례

- 실선 = 신규 벽
- 파선 = 기존 벽
- ○ = 콘센트
- ✕ = 조명
- SD/CO = 감지기
- P1..P4 = 관로

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LOT 7, CHAPPAREL LANDING DIVISION 2 · Tax-ID 4038-00-00700

AFH 전환 (기존 3-car garage) - 허가 제출용 스키매틱

원안: Canarch Designs (Caleb Mongare) - A01.3 / A01.4

Project No. CMN/25/122 Date OCT 2026

SHEET G-02 범례 및 시트 목록

이 도면은 A01.3/A01.4 기반 스키매틱입니다.
시공 전 실측 확인 필요. owner-builder 제출, 서명 불요.

항목	값	출처
주소	3571 Cassie Drive SW, Tumwater, WA 98512	카운티
Parcel / Tax-ID	40380000700 / 4038-00-00700	카운티 세무기록
대지면적	9,411 SF (0.216 ac)	카운티 기록
대지면적(측량)	9,357 SF	측량도/카운티
피복률	37.6 %	측량도
Zoning	R4R (측량도 표기 - 시 확인 필요)	측량도
준공연도	2016	카운티 기록
연면적	2,621 living / 3,411 gross	카운티 기록
1층 면적	1,160 SF	카운티 기록
침실/욕실	4 bd / 2 full + 1 half (MLS 2.5)	카운티 기록
기초	Concrete	카운티 기록
지붕/외벽	Gable / Composition Shingle / Wood Siding	카운티 기록
난방	Heat pump + gas fireplace (1)	카운티 기록
온수기	Gas, 기존 위치 차고 내 (전환 시 이전 필요)	MLS
주차/차고	Attached built-in, 790 SF (전환 대상)	Zillow + 카운티 assessor
상하수도	PSE 전기 / City of Tumwater 상·하수도	카운티 assessor
HOA	$900/yr (Chapparel Landing)	MLS/Zillow
과세가/연세액	$0 / $0	카운티 세무기록
측량도	L2-25T2-3R · J. Kerzie Hoyles LLC (253-371-6400) · 7/30/15 · City received 2016-03-02	측량도/카운티

주의: Zillow/MLS 값은 참고용이며, 허가 신청에는 카운티 assessor 기록과 실측이 우선합니다.

기존 온수기가 가스식이고 차고 내부에 설치되어 있어, 차고를 거주공간으로 전환할 경우

이전 설치 또는 연소공기·격리 조치가 필요합니다 (IRC M1307 / G2408 검토).

3571 Cassie Drive SW | Tumwater, WA 98512

LOT 7, CHAPPAREL LANDING DIVISION 2 · Tax-ID 4038-00-00700

AFH 전환 (기존 3-car garage) - 허가 제출용 스키매틱

원안: Canarch Designs (Caleb Mongare) - A01.3 / A01.4

Project No. CMN/25/122 Date OCT 2026

SHEET G-03 부동산·건물 정보 (public record)

이 도면은 A01.3/A01.4 기반 스키매틱입니다.
시공 전 실측 확인 필요. owner-builder 제출, 서명 불요.

Permit type	Residential Alteration / Change of Use (AFH)
Project address	3571 Cassie Drive SW, Tumwater, WA 98512
Legal	SECTION 28 TOWNSHIP 18 RANGE 2W QUARTER NE SE & SE SE PLAT CHAPPAREL LANDING DIVISION 2 LT 7 DOCUMENT 4490930
Parcel / Tax-ID	40380000700 / 4038-00-00700
Zoning	R4R (측량도 표기 - 시 확인 필요)
Lot / coverage	9,411 SF · 37.6% (측량)
Existing ground floor	1,160 SF of 2,621 SF living
Scope	기존 3-car garage → AFH 거주구역 전환 (침실 3 + 통합욕실)
Existing use	3-car garage (non-habitable)
Proposed use	AFH 거주공간 - 침실 3 + 통합욕실 1
Area changed	790 SF (garage) → 736 SF (habitable)
Construction value	$ (건축주 기재)
Applicant / Owner	(성명·연락처 기재)
Contractor	(LIC# 기재)
Owner-builder	본인 소유·직접 시공 (WA RCW 18.27.090 면제)
Project no.	CMN/25/122
Code year	2021 IRC + WA amendments, 2021 WSEC

서명: 건축주 _____ 날짜 _____

(이 시트는 신청 양식 항목을 정리한 것으로, 시 공식 양식 제출이 별도로 필요합니다.)

3571 Cassie Drive SW | Tumwater, WA 98512

LOT 7, CHAPPAREL LANDING DIVISION 2 · Tax-ID 4038-00-00700

AFH 전환 (기존 3-car garage) - 허가 제출용 스키매틱

원안: Canarch Designs (Caleb Mongare) - A01.3 / A01.4

Project No. CMN/25/122 Date OCT 2026

SHEET G-04 허가 신청 데이터 - City of Tumwater

이 도면은 A01.3/A01.4 기반 스키매틱입니다.
시공 전 실측 확인 필요. owner-builder 제출, 서명 불요.

항목	본 세트 위치	상태
허가 신청서	G-04 + 시 공식 양식	작성 필요
대지/주차 계획	C-01	포함(개념)
기존/제안 평면	A-01, A-02, A-05	포함
면적 산출	A-12	포함
입면/단면	A-04, A-13, A-14	개념 수준
구조 계획	S-01, S-02, S-03	EOR 검토 필요
기계/전기/배관	M-01~03, E-01~03, P-02~04	설계자 확정 필요
소방(연기/CO)	LS-01	포함
이그레스	LS-02	포함
에너지(WSEC)	EN-01, EN-02, EN-03	분석가 검토 필요
문/창 일람표	A-09, A-10	포함
자재 목록	자재목록.csv	예비
AFH 라이선스 서류	(DSHS 별도)	별도 제출
Owner-builder 선언서	G-04 + 시 양식	첨부

1. 상태가 '필요'인 항목은 면허 보유자 또는 시 공식 양식으로 보완해야 합니다.

2. 본 세트는 예비 도면이며, 날인 없는 제출은 반려됩니다.

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LOT 7, CHAPPAREL LANDING DIVISION 2 · Tax-ID 4038-00-00700

AFH 전환 (기존 3-car garage) - 허가 제출용 스키매틱

원안: Canarch Designs (Caleb Mongare) - A01.3 / A01.4

Project No. CMN/25/122 Date OCT 2026

SHEET G-05 제출 체크리스트

이 도면은 A01.3/A01.4 기반 스키매틱입니다.
시공 전 실측 확인 필요. owner-builder 제출, 서명 불요.

#	항목	출처 간 차이	처리
1	Lot 번호	카운티 LT 7 / 측량표지 LOT 1	카운티·시 확인
2	대지면적	카운티 9,411 / 측량 9,357	피복률은 측량 기준
3	Zoning	측량표기 R4R (현행 조례 확인)	시 조닝 확인
4	기초	MLS Block / 카운티 Concrete	현장 확인
5	1층 면적	카운티 1,160 / A01.4 736 SF	A01.4 기준 설계
6	가스 온수기	차고 내부 설치 (MLS)	이전/격리 계획
7	AFH 라이선스	건축 허가와 별개 (DSHS)	별도 진행
8	차고도어	유지 + 내부 스킨월 (A01.4)	구조 영향 없음

이 표는 제출 전 해소해야 할 상충 항목 기록입니다.

가스 온수기가 차고 내부에 설치 — 거주공간 전환 시 이전 또는 연소공기·격리 조치 필요 (IRC M1307 / G2408)

확인 전까지 도면의 legal/zoning 표기는 카운티 기록을 따릅니다.

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LOT 7, CHAPPAREL LANDING DIVISION 2 · Tax-ID 4038-00-00700

AFH 전환 (기존 3-car garage) - 허가 제출용 스키매틱

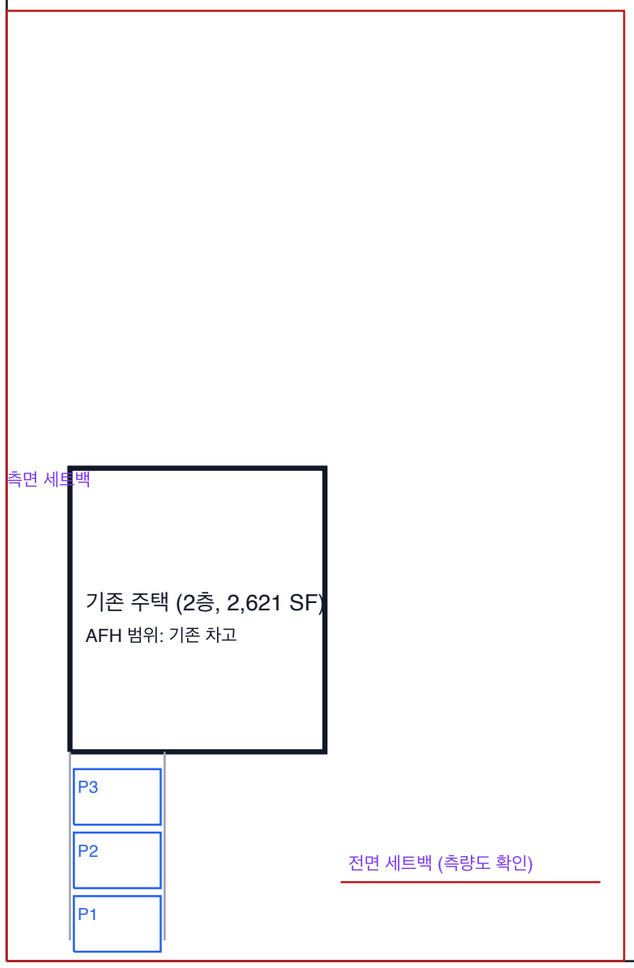
원안: Canarch Designs (Caleb Mongare) - A01.3 / A01.4

Project No. CMN/25/122 Date OCT 2026

SHEET G-06 조정(Discrepancy) 로그

이 도면은 A01.3/A01.4 기반 스키매틱입니다.
시공 전 실측 확인 필요. owner-builder 제출, 서명 불요.

LOT 9,357 SF (카운티) · 피복률 37.6% · 1:200



기존 진입로 / 주차 3면 (유지)

1. 근거: 측량도 L2-25T2-3R (2015, J. Kerzie Hoyles LLC), 시 접수 2016-03-02.
2. 측량도 표지: 대지 9,357 SF, 구조물 2,824 SF, 진입로 1,706 SF, 피복률 37.6%.
3. 카운티 세무기록: 대지 9,411 SF, LT 7. 측량도 표지는 LOT 1 → 번호 상이(확인 필요).
4. Zoning은 측량도 표기(R4R)를 기재. 현행 조례 확인 필요.
5. 주차 3면 유지. 공사 범위는 건물 내부로 대지 형질 변경 없음.

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LOT 7, CHAPPAREL LANDING DIVISION 2 · Tax-ID 4038-00-00700

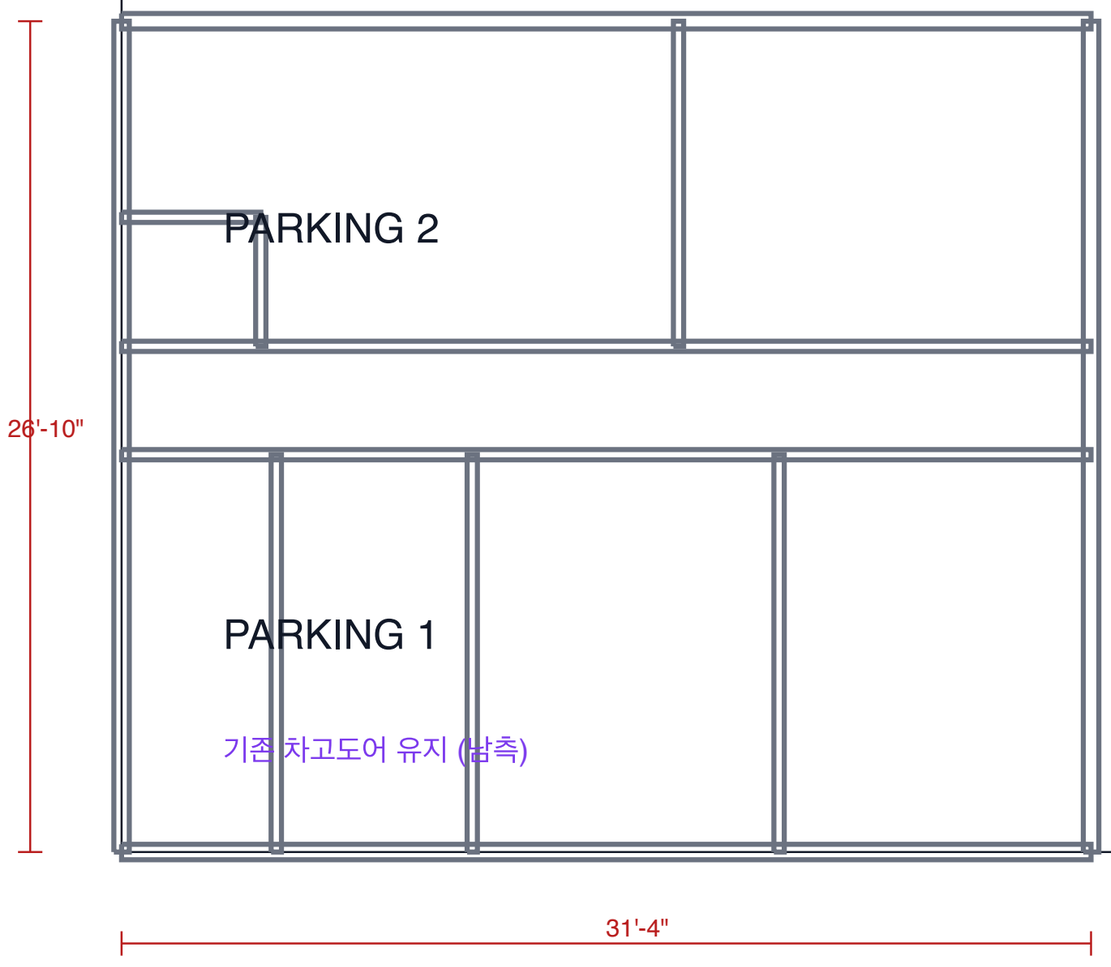
AFH 전환 (기존 3-car garage) - 허가 제출용 스키매틱

원안: Canarch Designs (Caleb Mongare) - A01.3 / A01.4

Project No. CMN/25/122 Date OCT 2026

SHEET C-01 대지·주차 계획 (1:200)

이 도면은 A01.3/A01.4 기반 스키매틱입니다.
시공 전 실측 확인 필요. owner-builder 제출, 서명 불요.



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AFH 전환 (기존 3-car garage) - 허가 제출용 스키매틱

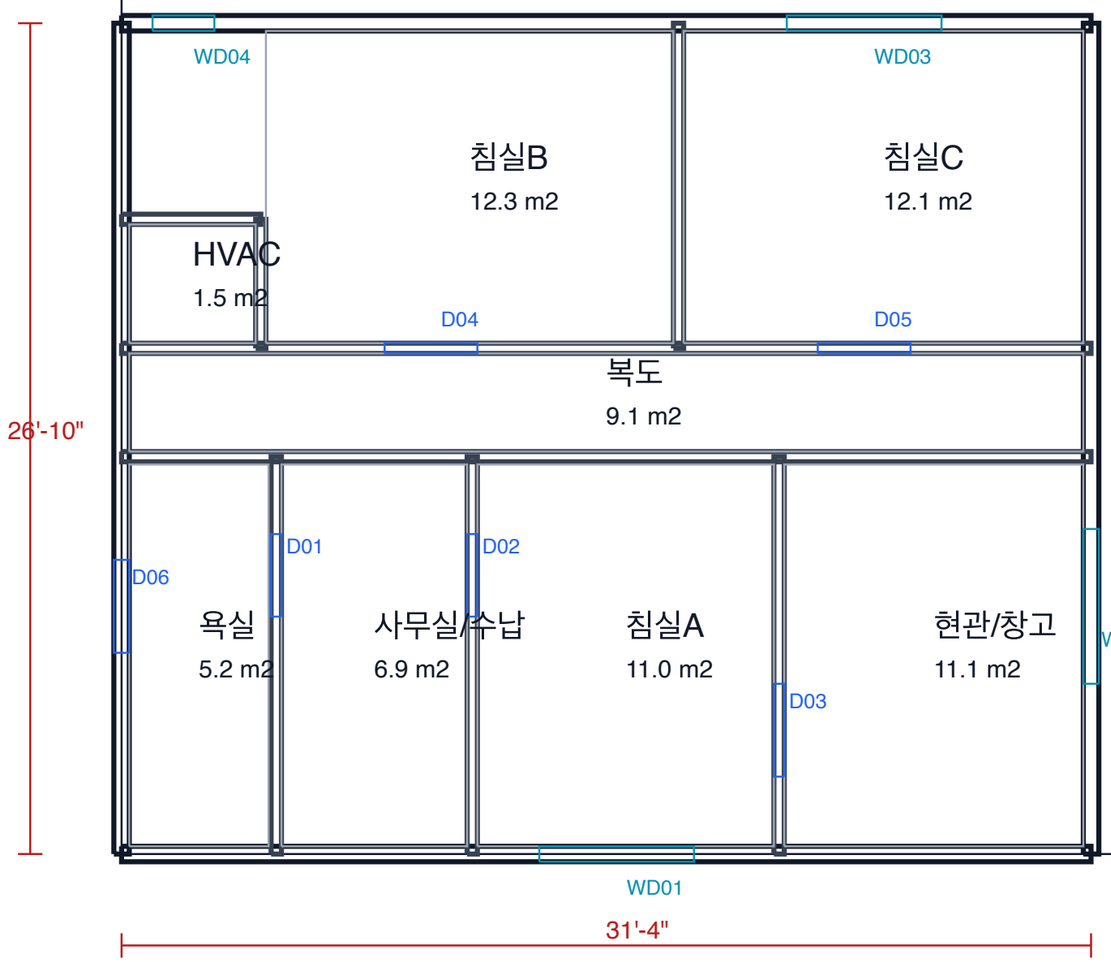
원안: Canarch Designs (Caleb Mongare) - A01.3 / A01.4

Project No. CMN/25/122 Date OCT 2026

SHEET A-01 기존 평면도

이 도면은 A01.3/A01.4 기반 스키매틱입니다.
시공 전 실측 확인 필요. owner-builder 제출, 서명 불요.

CEILING 10'-5" 내부면적 합계 69.0 m2



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AFH 전환 (기존 3-car garage) - 허가 제출용 스키매틱

원안: Canarch Designs (Caleb Mongare) - A01.3 / A01.4

Project No. CMN/25/122 Date OCT 2026

SHEET A-02 제안 평면도

이 도면은 A01.3/A01.4 기반 스키매틱입니다.
시공 전 실측 확인 필요. owner-builder 제출, 서명 불요.

실 면적표

실 이름	폭	깊이	m2	SF
욕실	4'-6"	12'-4"	5.16	55
사무실/수납	6'-0"	12'-4"	6.87	74
침실A	9'-7"	12'-4"	10.98	118
현관/창고	9'-8"	12'-4"	11.08	119
복도	30'-10"	3'-2"	9.07	98
HVAC	4'-1"	3'-10"	1.45	16
침실B	13'-2"	10'-1"	12.33	133
침실C	12'-11"	10'-1"	12.10	130

합계 69.05 m2 (743 SF)

문 일람표

문	폭 x 높이
D01 욕실	2'-8" x 6'-8"
D02 사무실	2'-8" x 6'-8"
D03 침실A	3'-0" x 6'-8"
D04 침실B	3'-0" x 6'-8"
D05 침실C	3'-0" x 6'-8"
D06 기존주택 연결	3'-0" x 6'-8"

창 일람표

창	폭 x 높이	창턱
WD01 침실A 이그레스	5'-0" x 4'-0"	3'-4"
WD02 침실B 이그레스	5'-0" x 4'-0"	3'-4"
WD03 침실C 이그레스	5'-0" x 4'-0"	3'-4"
WD04 욕실 창	2'-0" x 2'-0"	5'-0"

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AFH 전환 (기존 3-car garage) - 허가 제출용 스키매틱

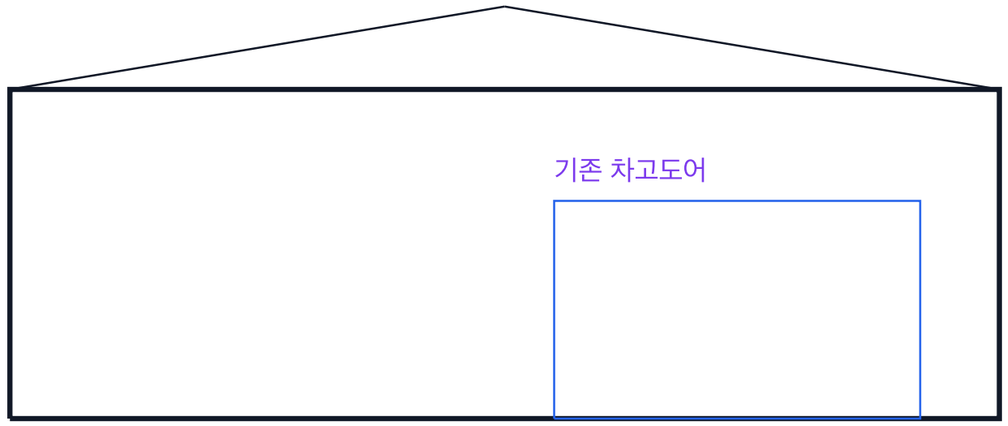
원안: Canarch Designs (Caleb Mongare) - A01.3 / A01.4

Project No. CMN/25/122 Date OCT 2026

SHEET A-03 실 면적표 / 문·창 일람표

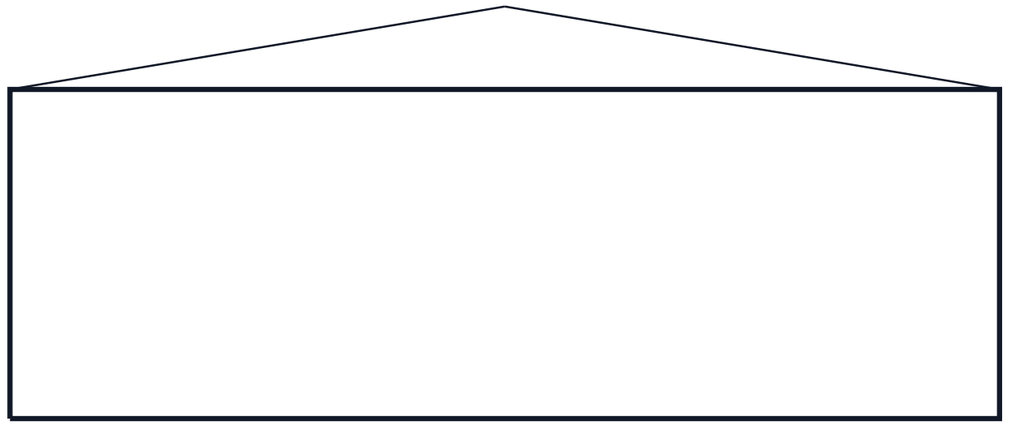
이 도면은 A01.3/A01.4 기반 스키매틱입니다.
시공 전 실측 확인 필요. owner-builder 제출, 서명 불요.

남측 SOUTH



CEILING 10'-5"

북측 NORTH



CEILING 10'-5"

동측 EAST / 서측 WEST (개념)



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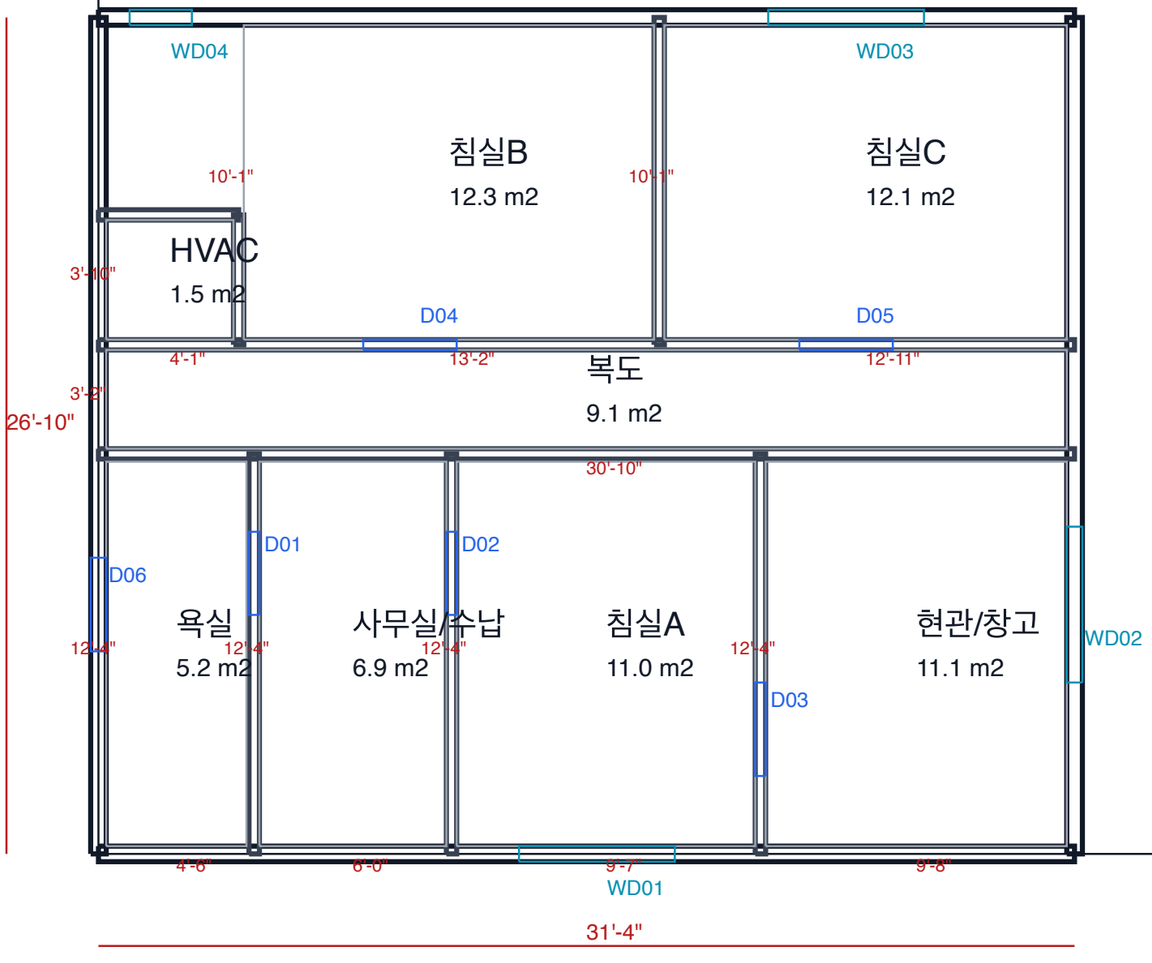
AFH 전환 (기존 3-car garage) - 허가 제출용 스키매틱

원안: Canarch Designs (Caleb Mongare) - A01.3 / A01.4

Project No. CMN/25/122 Date OCT 2026

SHEET A-04 입면도 (개념)

이 도면은 A01.3/A01.4 기반 스키매틱입니다.
시공 전 실측 확인 필요. owner-builder 제출, 서명 불요.



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AFH 전환 (기존 3-car garage) - 허가 제출용 스키매틱

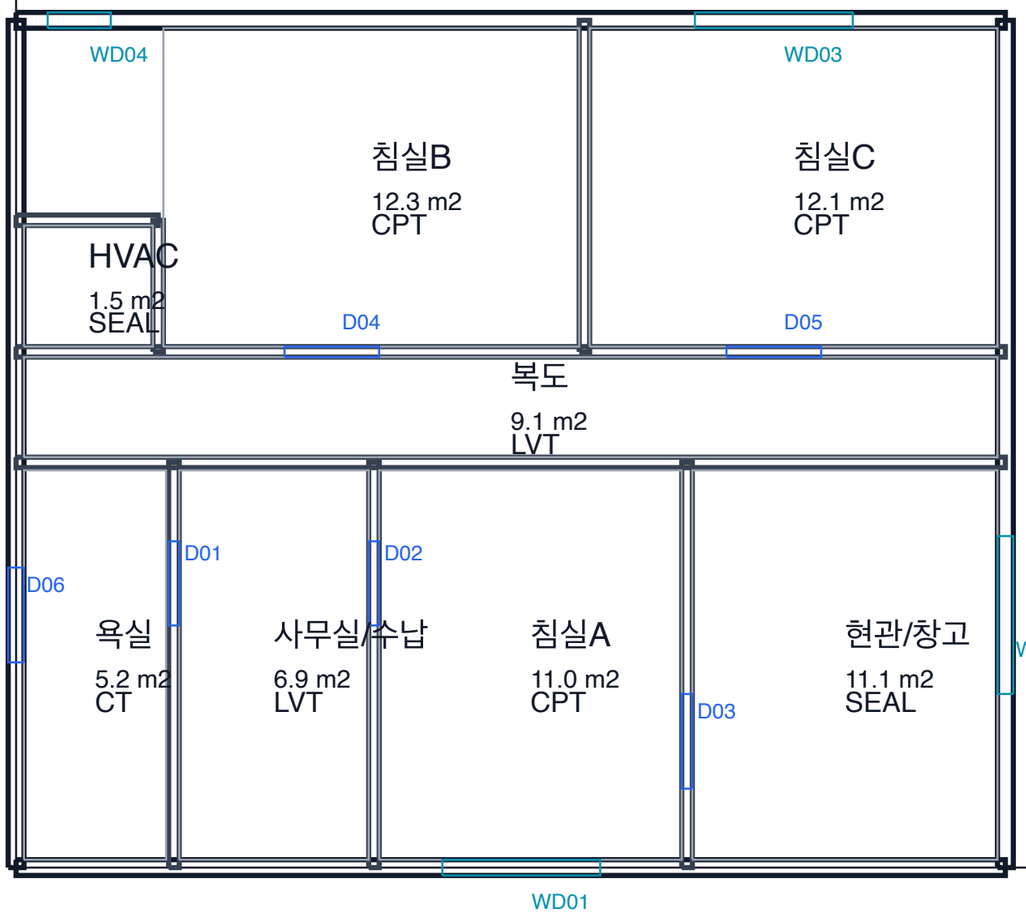
원안: Canarch Designs (Caleb Mongare) - A01.3 / A01.4

Project No. CMN/25/122 Date OCT 2026

SHEET A-05 치수 평면도

이 도면은 A01.3/A01.4 기반 스키매틱입니다.
시공 전 실측 확인 필요. owner-builder 제출, 서명 불요.

코드	마감	적용
CPT	카펫 타일	침실
LVT	비닐 타일	복도/사무실
CT	세라믹 타일	욕실
SEAL	실링	수납/기계
PT	페인트	전체 벽/천장



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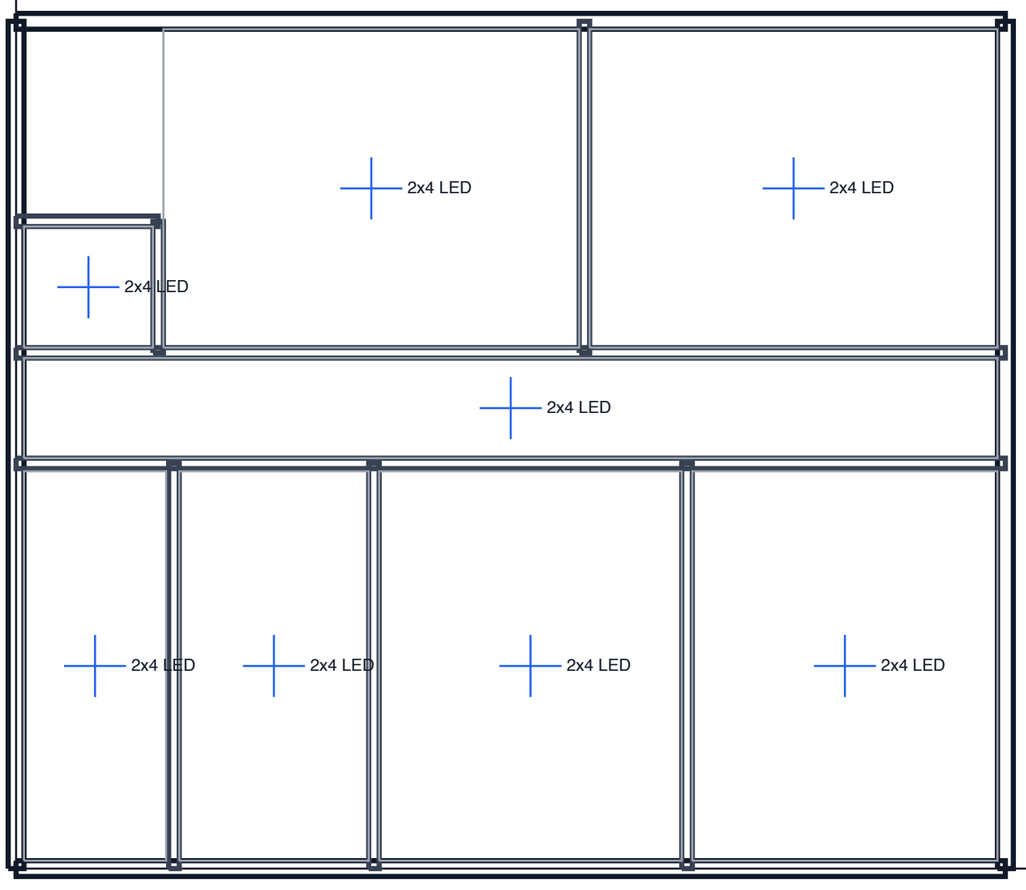
AFH 전환 (기존 3-car garage) - 허가 제출용 스키매틱

원안: Canarch Designs (Caleb Mongare) - A01.3 / A01.4

Project No. CMN/25/122 Date OCT 2026

SHEET A-06 마감 평면도

이 도면은 A01.3/A01.4 기반 스키매틱입니다.
시공 전 실측 확인 필요. owner-builder 제출, 서명 불요.



1. 천장고: 10'-5" (설계 기준), 마감 후 실내고 확인
2. 신규 파티션은 천장까지 (방음·차음)
3. LED 2x4 패널, 조광 가능, 난연 등급
4. 욕실: 방습 천장재 + 배기팬(EF-1)

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AFH 전환 (기존 3-car garage) - 허가 제출용 스키매틱

원안: Canarch Designs (Caleb Mongare) - A01.3 / A01.4

Project No. CMN/25/122 Date OCT 2026

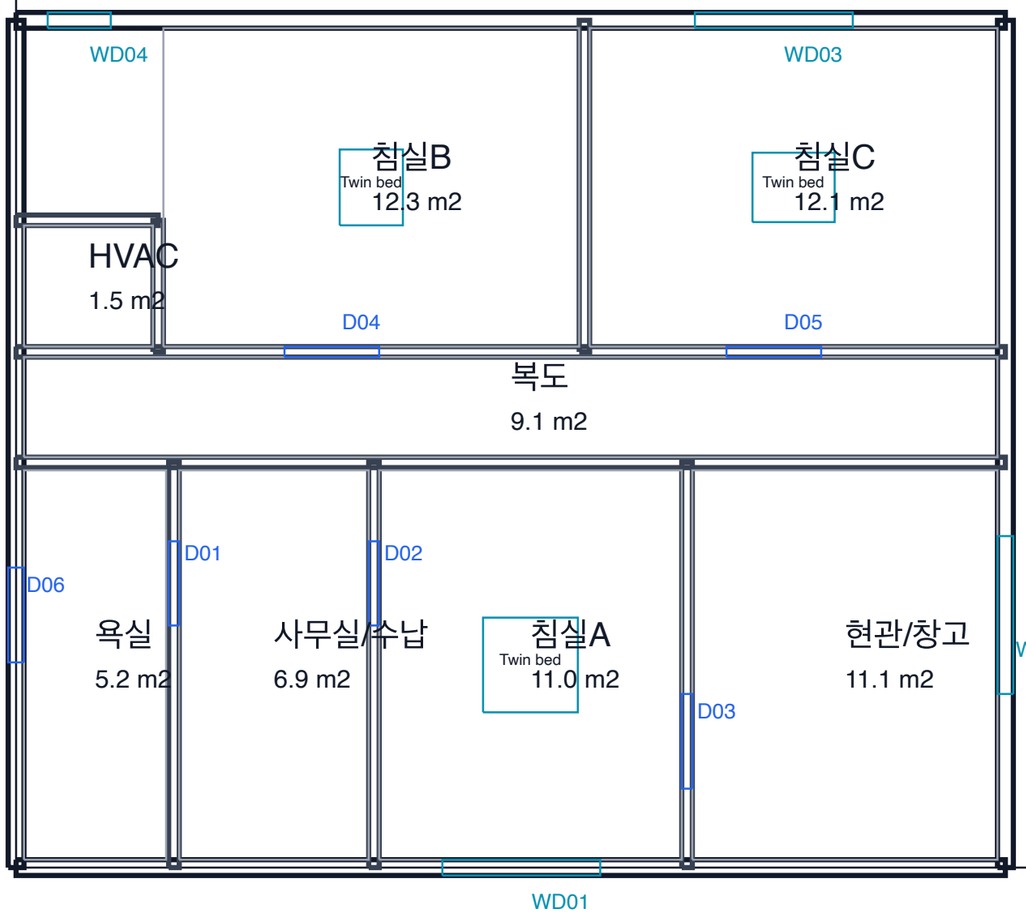
SHEET A-07 반사천장 평면도 (RCP)

이 도면은 A01.3/A01.4 기반 스키매틱입니다.
시공 전 실측 확인 필요. owner-builder 제출, 서명 불요.

AFH 가구 기준: 침실당 침대 1 + 옷장 1 (DSHS WAC 388-76 검토)

욕실: 접근 가능한 샤워(의자·잡이) 확보 권장

복도 유효폭 36in 이상 유지 (가구 배치 후 확인)



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AFH 전환 (기존 3-car garage) - 허가 제출용 스키매틱

원안: Canarch Designs (Caleb Mongare) - A01.3 / A01.4

Project No. CMN/25/122 Date OCT 2026

SHEET A-08 가구·설비 배치도

이 도면은 A01.3/A01.4 기반 스키매틱입니다.
시공 전 실측 확인 필요. owner-builder 제출, 서명 불요.

MARK	위치	폭x높이	타입
D01	W05	2'-8"x6'-8"	HM/단열
D02	W06	2'-8"x6'-8"	HM/단열
D03	W07	3'-0"x6'-8"	HM/단열
D04	W09	3'-0"x6'-8"	HM/단열
D05	W09	3'-0"x6'-8"	HM/단열
D06	W03	3'-0"x6'-8"	HM/단열

내부문: 2'-8" / 3'-0" (AFH 통과폭 WAC 388-76-10715 검토)

이그레스 문: 3'-0" x 6'-8", 32in 순폭 확인

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AFH 전환 (기존 3-car garage) - 허가 제출용 스키매틱

원안: Canarch Designs (Caleb Mongare) - A01.3 / A01.4

Project No. CMN/25/122 Date OCT 2026

SHEET A-09 문 일람표

이 도면은 A01.3/A01.4 기반 스키매틱입니다.
시공 전 실측 확인 필요. owner-builder 제출, 서명 불요.

MARK	위치	크기	창턱	유리
WD01	W01	5'-0"x4'-0"	3'-4"	Insul. IG
WD02	W04	5'-0"x4'-0"	3'-4"	Insul. IG
WD03	W02	5'-0"x4'-0"	3'-4"	Insul. IG
WD04	W02	2'-0"x2'-0"	5'-0"	Insul. IG

이그레스 창: 순개구부 5.7 sqft, 최소 20in x 24in, 바닥 44in 이하 (IRC R310)

창호 성능은 EN-02 참조 (U-0.30 이하)

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AFH 전환 (기존 3-car garage) - 허가 제출용 스키매틱

원안: Canarch Designs (Caleb Mongare) - A01.3 / A01.4

Project No. CMN/25/122 Date OCT 2026

SHEET A-10 창 일람표

이 도면은 A01.3/A01.4 기반 스키매틱입니다.
시공 전 실측 확인 필요. owner-builder 제출, 서명 불요.

실	바닥	벽	천장
침실A/B/C	CPT	PT + GWB	PT + GWB
욕실	CT	CT + 방수	방습 PT
복도	LVT	PT + GWB	PT + GWB
사무실/수납	LVT	PT + GWB	PT + GWB
HVAC	SEAL	GWB	GWB

욕실 방수: 바닥 전체 + 벽 1.2m 이상, 샤워부 1.8m

모든 GWB 이음부 테이프·퍼티, 나사 피치 12in

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AFH 전환 (기존 3-car garage) - 허가 제출용 스키매틱

원안: Canarch Designs (Caleb Mongare) - A01.3 / A01.4

Project No. CMN/25/122 Date OCT 2026

SHEET A-11 마감 일람표

이 도면은 A01.3/A01.4 기반 스키매틱입니다.
시공 전 실측 확인 필요. owner-builder 제출, 서명 불요.

항목	m2	SF
Gross (외곽)	78.1	841
Net (실 합계)	69.0	743
차이(벽체 등)	9.1	98
욕실	5.16	55
사무실/수납	6.87	74
침실A	10.98	118
현관/창고	11.08	119
복도	9.07	98
HVAC	1.45	16
침실B	12.33	133
침실C	12.10	130

설계 기준 내부면적 736 SF 와 대조 필요

AFH 침실 최소 면적 기준은 DSHS 규정 확인

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AFH 전환 (기존 3-car garage) - 허가 제출용 스키매틱

원안: Canarch Designs (Caleb Mongare) - A01.3 / A01.4

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SHEET A-12 면적 산출 (Net/Gross)

이 도면은 A01.3/A01.4 기반 스키매틱입니다.
시공 전 실측 확인 필요. owner-builder 제출, 서명 불요.

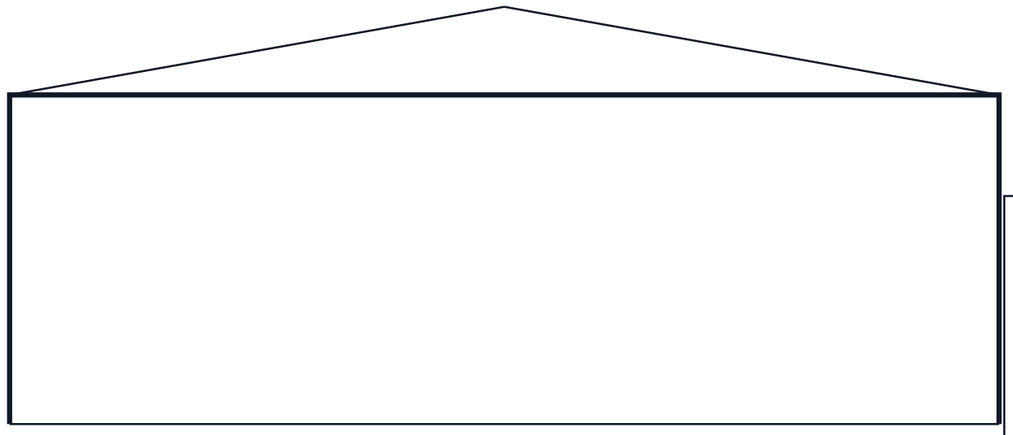
1. 천장고 10'-5" (설계 기준)

2. 지붕 구조는 기존 유지

3. 천장 단열 R-49 (EN-01)

4. 벽체 구성은 A-15 상세 참조

CEILING 10'-5"



단면 위치: y = 14.0 ft

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AFH 전환 (기존 3-car garage) - 허가 제출용 스키매틱

원안: Canarch Designs (Caleb Mongare) - A01.3 / A01.4

Project No. CMN/25/122 Date OCT 2026

SHEET A-13 단면 A-A

이 도면은 A01.3/A01.4 기반 스키매틱입니다.
시공 전 실측 확인 필요. owner-builder 제출, 서명 불요.

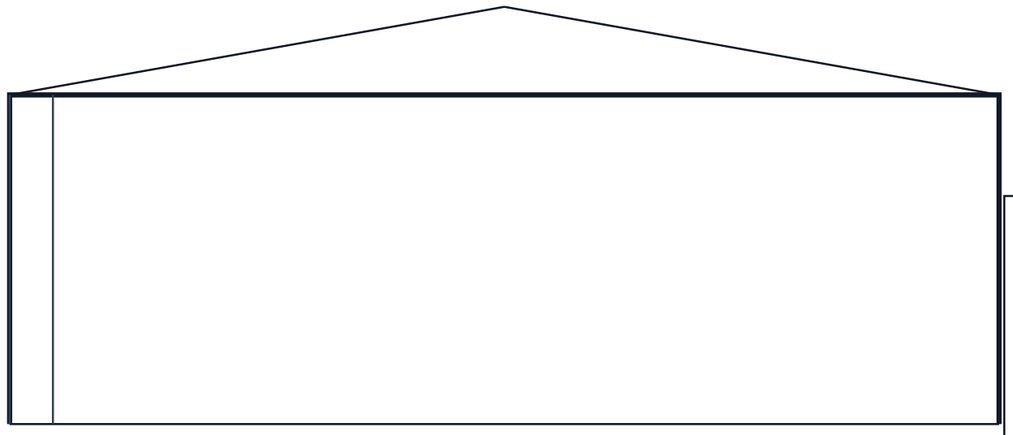
1. 천장고 10'-5" (설계 기준)

2. 지붕 구조는 기존 유지

3. 천장 단열 R-49 (EN-01)

4. 벽체 구성은 A-15 상세 참조

CEILING 10'-5"



단면 위치: y = 21.0 ft

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AFH 전환 (기존 3-car garage) - 허가 제출용 스키매틱

원안: Canarch Designs (Caleb Mongare) - A01.3 / A01.4

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SHEET A-14 단면 B-B

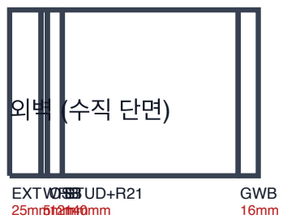
이 도면은 A01.3/A01.4 기반 스키매틱입니다.
시공 전 실측 확인 필요. owner-builder 제출, 서명 불요.

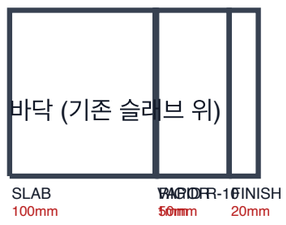
1. 외벽 R-21 이상 (WSEC), 외장 단열 검토

2. 기밀: 모든 관통부 실란트

3. 방수: 욕실 바닥·벽체 방수층 상향 1.2m

4. 슬래브 단열로 바닥 냉기 차단





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AFH 전환 (기존 3-car garage) - 허가 제출용 스키매틱

원안: Canarch Designs (Caleb Mongare) - A01.3 / A01.4

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SHEET A-15 외벽·바닥 상세 (전형)

이 도면은 A01.3/A01.4 기반 스키매틱입니다.
시공 전 실측 확인 필요. owner-builder 제출, 서명 불요.

1. 접이식 샤워 의자·잡이 설치 권장 (AFH 접근성)
2. 배수 2in, 구배 1/4in/ft
3. 환기 EF-1 80 CFM (M-02)
4. 미끄럼 방지 타일 (COF 0.6 이상)



방수층 (바닥+벽 1.2m, 샤워 1.8m)

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AFH 전환 (기존 3-car garage) - 허가 제출용 스키매틱

원안: Canarch Designs (Caleb Mongare) - A01.3 / A01.4

Project No. CMN/25/122 Date OCT 2026

SHEET A-16 욕실 상세 (방수·기구)

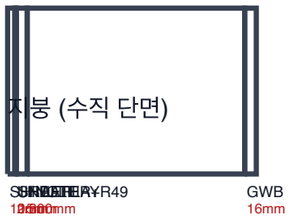
이 도면은 A01.3/A01.4 기반 스키매틱입니다.
시공 전 실측 확인 필요. owner-builder 제출, 서명 불요.

1. 기존 지붕 구조 유지, 신규 관통부 플래싱

2. 천장 단열 R-49 + 방습

3. 처마 통기 확보 (막힘 없도록)

4. 지붕 관통 배기 후드 역류방지





처마: 통기·드립에지·빗물받이

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AFH 전환 (기존 3-car garage) - 허가 제출용 스키매틱

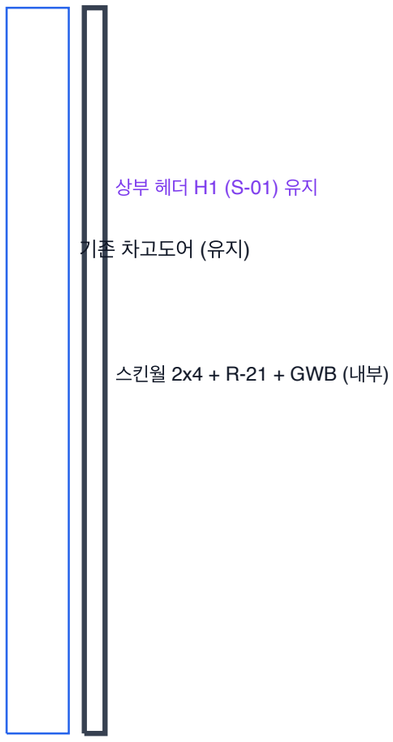
원안: Canarch Designs (Caleb Mongare) - A01.3 / A01.4

Project No. CMN/25/122 Date OCT 2026

SHEET A-17 지붕·처마 상세 (전형)

이 도면은 A01.3/A01.4 기반 스키매틱입니다.
시공 전 실측 확인 필요. owner-builder 제출, 서명 불요.

1. 도어는 구조적으로 유지하되 내부에서 단열·마감
2. 스킨월 하부 방수·기밀 (외부 습기 차단)
3. 도어 트랙·스프링 상태 점검 (기존 부재)
4. 필요 시 헤더 보강 — EOR 판단



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AFH 전환 (기존 3-car garage) - 허가 제출용 스키매틱

원안: Canarch Designs (Caleb Mongare) - A01.3 / A01.4

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SHEET A-18 차고도어 스킨월 상세

이 도면은 A01.3/A01.4 기반 스키매틱입니다.
시공 전 실측 확인 필요. owner-builder 제출, 서명 불요.

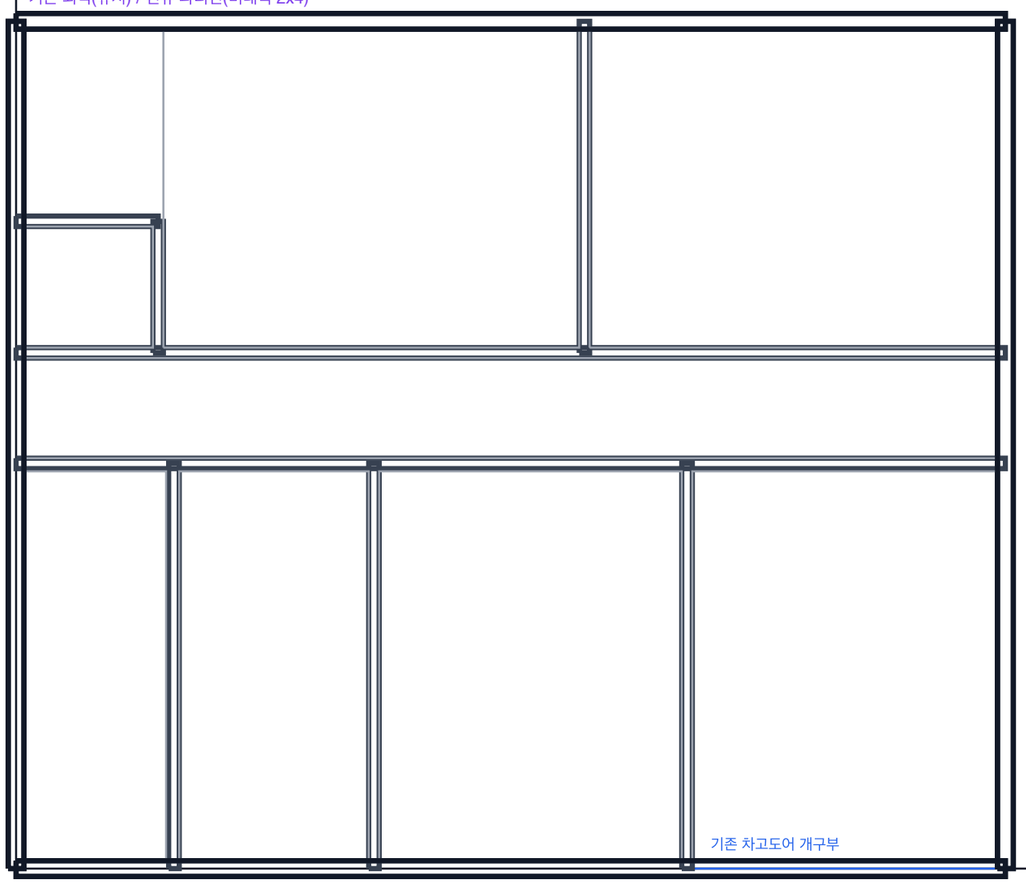
부재	위치	예비 단면	비고
헤더 H1	차고도어 개구부	2-1.75x11.875 LVL	EOR 확인
헤더 H2	W06 사무실/침실A	2x10 DF#2	비내력
스터드	신규 파티션	2x4 @16 o.c.	8ft
슬래브	기존 차고 바닥	기존 유지	레벨·균열
앵커	신규 벽 하부	1/2in @48in	슬래브 천공

예비 구조 계획입니다. 시 담당자 확인에 따라 구조기술사 서명 없이 제출하며, 요청 시 하중 검토를 보완합니다.

기존 슬래브 두께·철근 상태는 현장 확인 필요(거실 마감 하중).

신규 파티션은 비내력으로 계획, 개구부 신설 시 하중 경로 EOR 확인.

기존 외벽(유지) / 신규 파티션(비내력 2x4)



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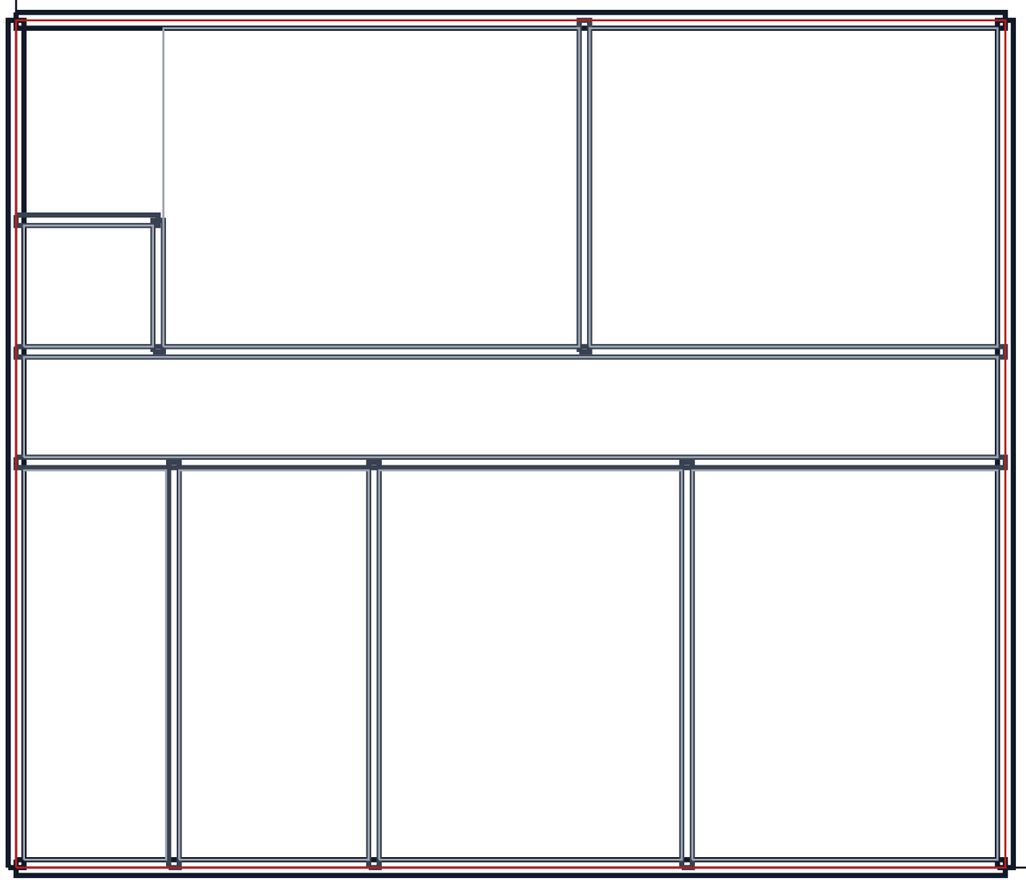
AFH 전환 (기존 3-car garage) - 허가 제출용 스키매틱

원안: Canarch Designs (Caleb Mongare) - A01.3 / A01.4

Project No. CMN/25/122 Date OCT 2026

SHEET S-01 구조 계획도 (예비)

이 도면은 A01.3/A01.4 기반 스키매틱입니다.
시공 전 실측 확인 필요. owner-builder 제출, 서명 불요.



1. 기존 차고 슬래브 유지 (두께·철근 상태 현장 확인)

2. 신규 파티션 하부: 1/2in 앵커 @48in, 슬래브 천공 후 에폭시 정착

3. 바닥 마감: 기존 슬래브 위 방습·단열·마감층 (상세 A-15 참조)

4. 배수 구배: 기존 바닥 레벨 확인 후 필요 시 자평 시멘트

5. 기초 신설 없음(기존 건물 내부 개조)

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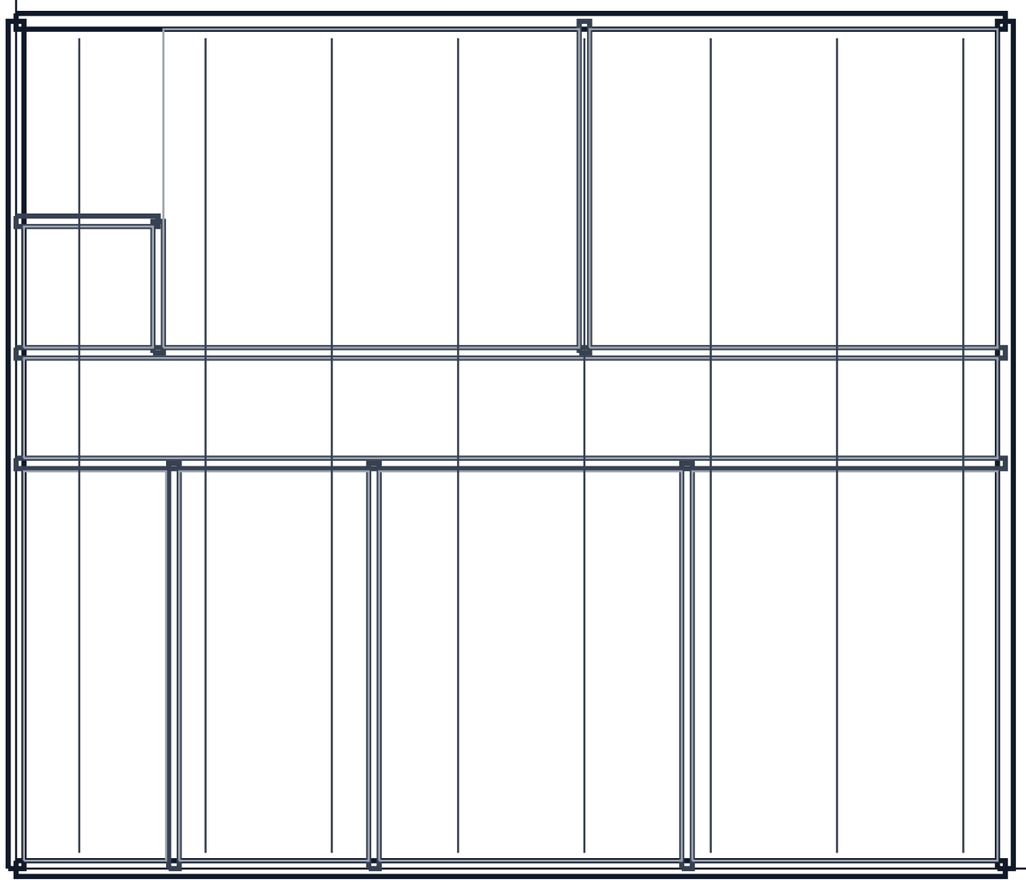
AFH 전환 (기존 3-car garage) - 허가 제출용 스키매틱

원안: Canarch Designs (Caleb Mongare) - A01.3 / A01.4

Project No. CMN/25/122 Date OCT 2026

SHEET S-02 기초·슬래브 계획 (예비)

이 도면은 A01.3/A01.4 기반 스키매틱입니다.
시공 전 실측 확인 필요. owner-builder 제출, 서명 불요.



1. 기존 지붕 구조 유지 (트러스/서까래 현장 확인)
2. 천장 장선: 기존 구조에 매달기, 신규 하중 없음
3. 신규 파티션 상부: 천장 장선 사이 블로킹
4. 천장 단열 R-49 (EN-01), 관통부 기밀 테이프
5. 지붕 하중 경로는 기존 외벽 유지 — EOR 확인

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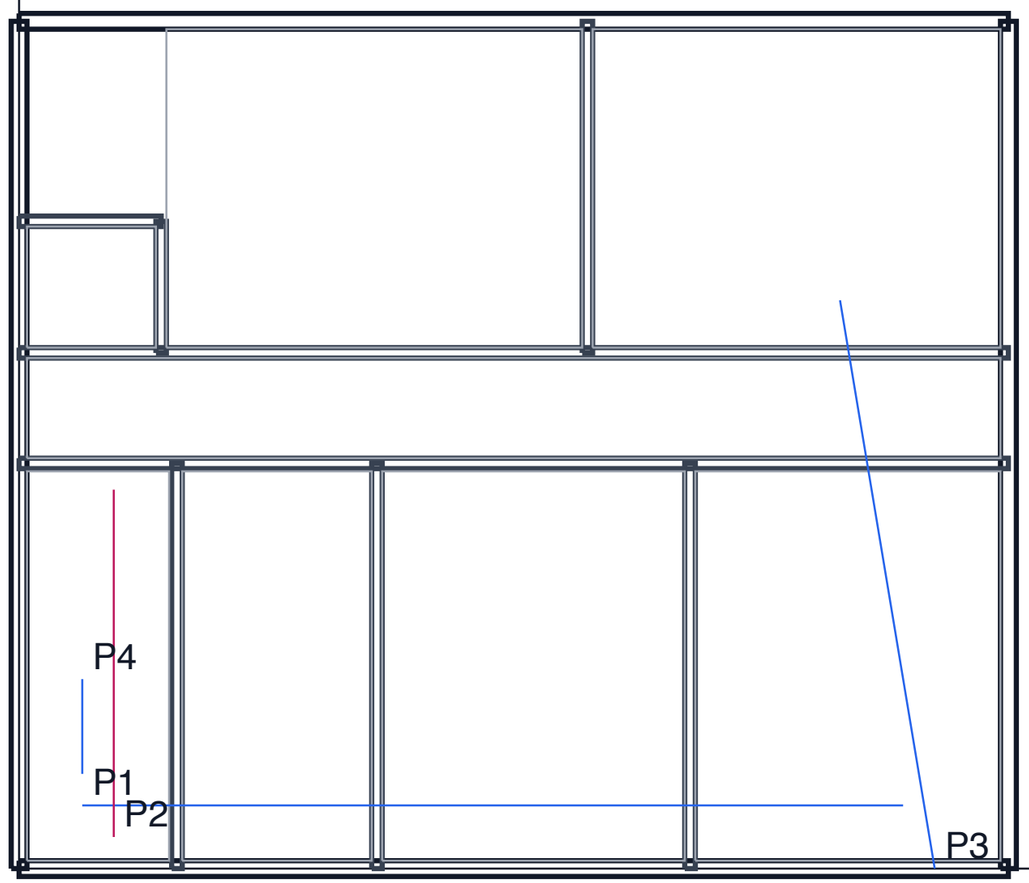
AFH 전환 (기존 3-car garage) - 허가 제출용 스키매틱

원안: Canarch Designs (Caleb Mongare) - A01.3 / A01.4

Project No. CMN/25/122 Date OCT 2026

SHEET S-03 지붕·천장장선 계획 (예비)

이 도면은 A01.3/A01.4 기반 스키매틱입니다.
시공 전 실측 확인 필요. owner-builder 제출, 서명 불요.



관로 일람표

기호	종류	관경	보온	구배
P1	급수	20 mm	20 mm	0.0 %
P2	배수	75 mm	0 mm	2.0 %
P3	냉매	16 mm	13 mm	0.0 %
P4	배기	100 mm	0 mm	0.0 %

EF-1 / EH-1: 욕실 배기팬 + 외벽 후드(역류방지). 관경/높이는 설계자 확인 필요.

CU-1: 실외 콘덴서 콘크리트 패드. 위치는 사이트/HOA 확인 후 확정.

DH-1: 드라이어 배관은 잠정(TBC). 별도 덕트, 백드래프트 댐퍼, 스크린 없음.

연기/CO 감지기: 상호연동 + 배터리 백업 (WAC R314.4 / IRC 314.4, R315.5).

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AFH 전환 (기존 3-car garage) - 허가 제출용 스키매틱

원안: Canarch Designs (Caleb Mongare) - A01.3 / A01.4

Project No. CMN/25/122 Date OCT 2026

SHEET M-01 설비 개념도

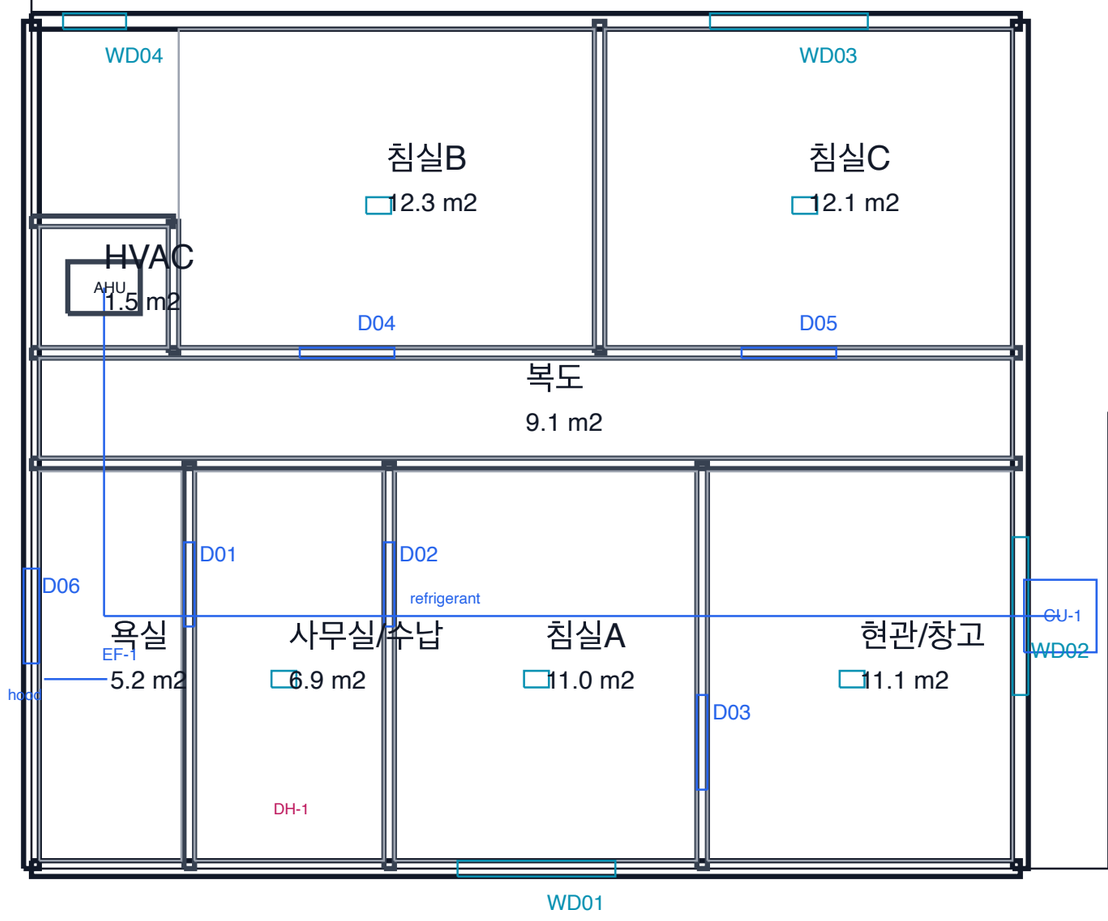
이 도면은 A01.3/A01.4 기반 스키매틱입니다.
시공 전 실측 확인 필요. owner-builder 제출, 서명 불요.

기호	설비	사양	비고
EF-1	욕실 배기팬	80 CFM	후드+역류방지
AHU	실내기	1.5 ton	냉매 R-454B
CU-1	실외기	1.5 ton	패드 이격 확인
DH-1	드라이어	4in 덕트	위치 TBC

장비 용량은 예비안. 최종 선정·경로는 기계설비 설계자 확정.

욕실 배기: 덕트 최단, 외벽 후드 높이는 설계자 확인.

전기 히트펌프 가정(연소기기 없음) — CO는 차고 인접 기준.



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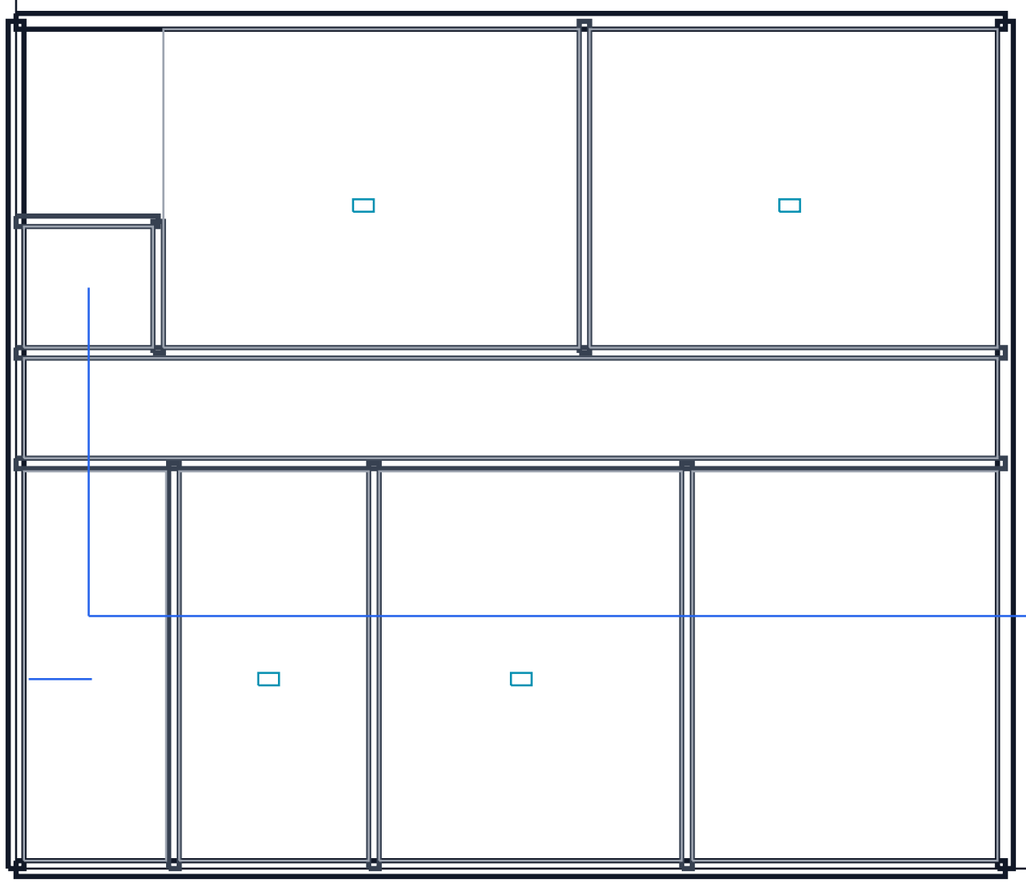
AFH 전환 (기존 3-car garage) - 허가 제출용 스키매틱

원안: Canarch Designs (Caleb Mongare) - A01.3 / A01.4

Project No. CMN/25/122 Date OCT 2026

SHEET M-02 기계 설비 평면

이 도면은 A01.3/A01.4 기반 스키매틱입니다.
시공 전 실측 확인 필요. owner-builder 제출, 서명 불요.



1. EF-1: 욕실 80 CFM, 외벽 후드
2. AHU → 공급 4개소 + 리턴 2개소
3. 덕트 방청·보온, 소음 25 NR 이하 목표
4. 드라이어(DH-1)는 별도 4in 덕트 (TBC)

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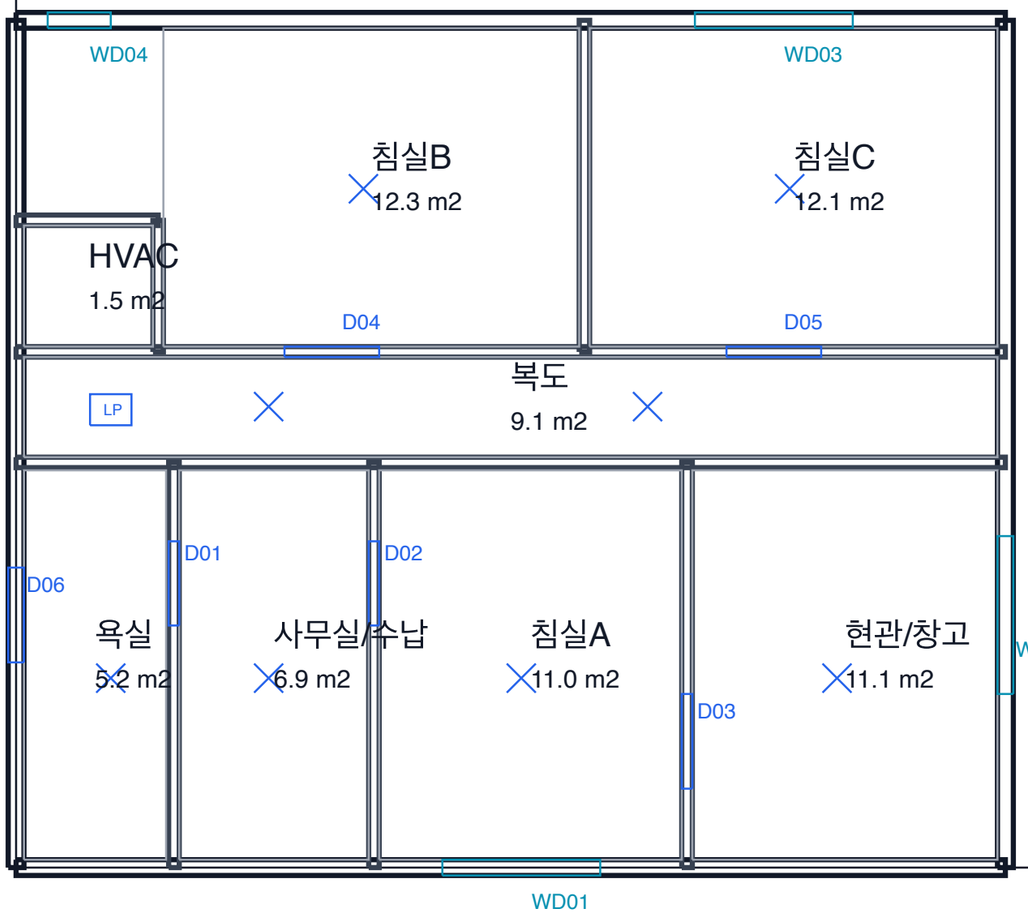
AFH 전환 (기존 3-car garage) - 허가 제출용 스키매틱

원안: Canarch Designs (Caleb Mongare) - A01.3 / A01.4

Project No. CMN/25/122 Date OCT 2026

SHEET M-03 덕트·환기 계통도

이 도면은 A01.3/A01.4 기반 스키매틱입니다.
시공 전 실측 확인 필요. owner-builder 제출, 서명 불요.



회로	용도	차단기	비고
C1	조명 (거실/복도)	15A	LED
C2	침실 조명·콘센트	20A	AFCI
C3	욕실	20A	GFCI
C4	사무실 콘센트	20A	AFCI
C5	AHU / CU-1	30A	전용
C6	온수기	30A	전용
C7	연기·CO 경보	15A	상호연동

○ 콘센트 (6ft 이내 간격, IRC E3901.2) / ✕ 조명

욕실 GFCI, 침실 AFCI (IRC E3902)

연기·CO는 상호연동 배선 + 배터리 백업

회로 구성은 예비안 — 전기설계자 확정 필요

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AFH 전환 (기존 3-car garage) - 허가 제출용 스키매틱

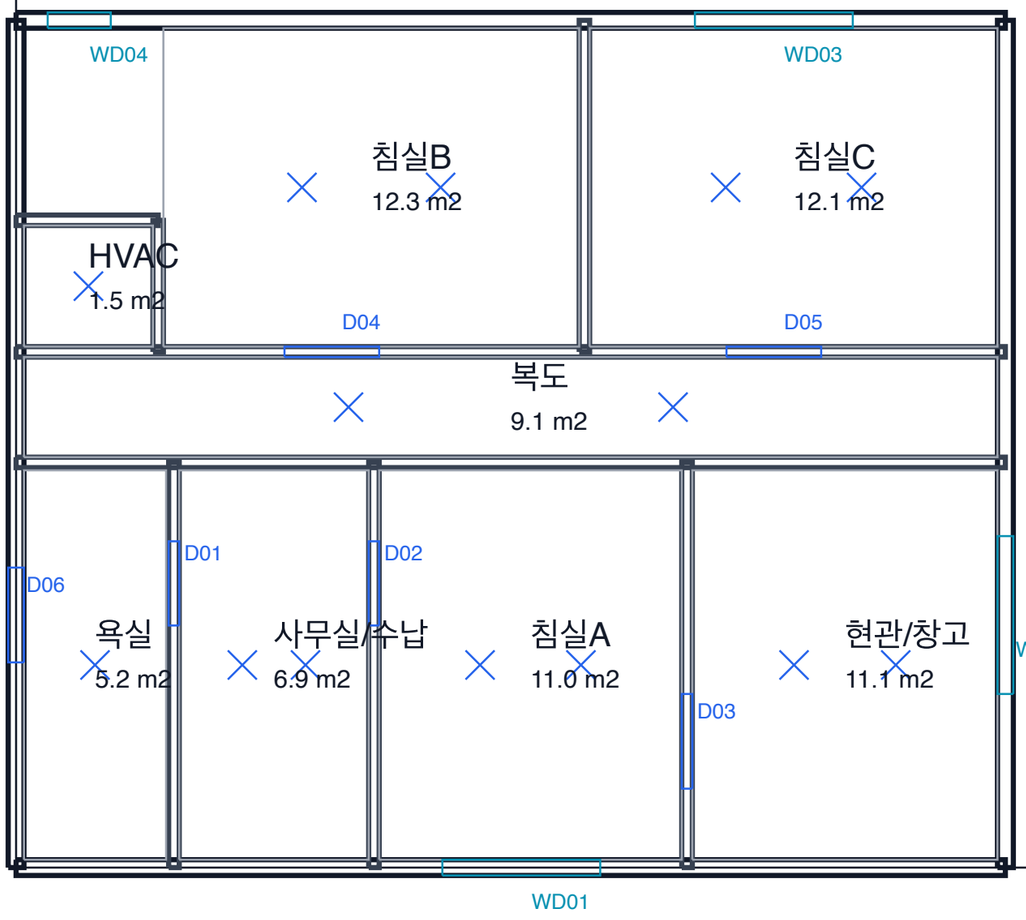
원안: Canarch Designs (Caleb Mongare) - A01.3 / A01.4

Project No. CMN/25/122 Date OCT 2026

SHEET E-01 전기 설비 평면

이 도면은 A01.3/A01.4 기반 스키매틱입니다.
시공 전 실측 확인 필요. owner-builder 제출, 서명 불요.

회로	부하	수량
C1	복도/거실 LED	6
C2	침실 LED	5
C3	욕실·방습	2
C4	사무실	2



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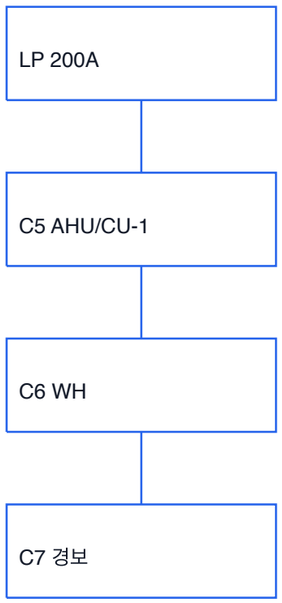
AFH 전환 (기존 3-car garage) - 허가 제출용 스키매틱

원안: Canarch Designs (Caleb Mongare) - A01.3 / A01.4

Project No. CMN/25/122 Date OCT 2026

SHEET E-02 전기 조명 평면

이 도면은 A01.3/A01.4 기반 스키매틱입니다.
시공 전 실측 확인 필요. owner-builder 제출, 서명 불요.



1. 분전반 200A (기존 용량 확인)
2. AHU/CU-1 30A 전용, 차단기·과전류 보호
3. 온수기 30A 전용
4. 경보 회로는 상호연동 + 배터리 백업
5. 결선도는 예비안 — 전기설계자 확정

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AFH 전환 (기존 3-car garage) - 허가 제출용 스키매틱

원안: Canarch Designs (Caleb Mongare) - A01.3 / A01.4

Project No. CMN/25/122 Date OCT 2026

SHEET E-03 전기 동력·장비 결선도

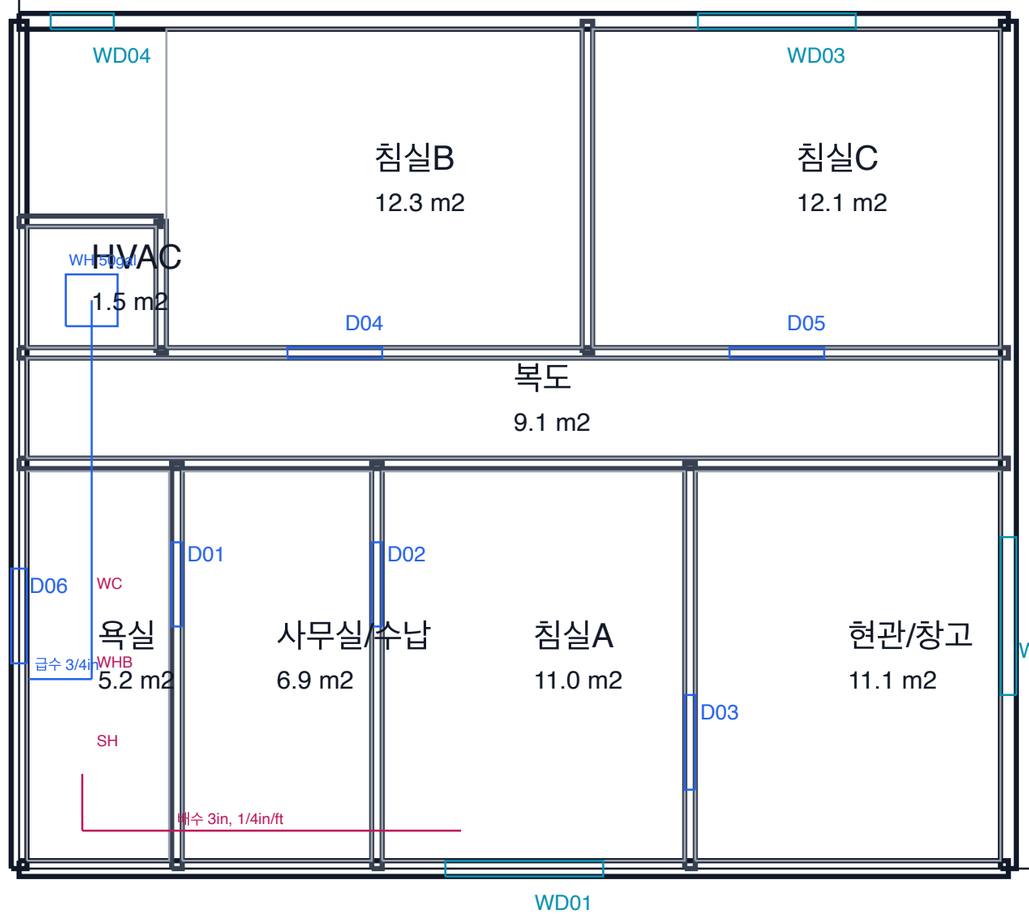
이 도면은 A01.3/A01.4 기반 스키매틱입니다.
시공 전 실측 확인 필요. owner-builder 제출, 서명 불요.

기구	수량	급수	배수
WC	1	1/2in	3in
세면기	1	3/8in	1-1/2in
샤워	1	1/2in	2in
온수기	1	3/4in	—

급수 3/4in → 온수기 → 욕실 분기(1/2in, 3/8in)

배수 3in 주관, 구배 1/4in/ft(2%), 통기 2in (IRC P3103)

관경·기구단위는 예비안. 배관설계자 확정 필요.



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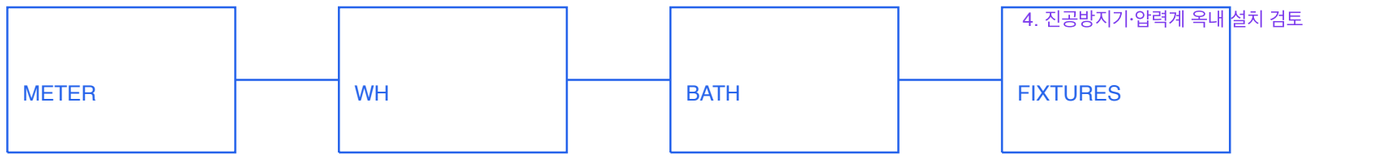
AFH 전환 (기존 3-car garage) - 허가 제출용 스키매틱

원안: Canarch Designs (Caleb Mongare) - A01.3 / A01.4

Project No. CMN/25/122 Date OCT 2026

SHEET P-02 급배수 평면

이 도면은 A01.3/A01.4 기반 스키매틱입니다.
시공 전 실측 확인 필요. owner-builder 제출, 서명 불요.



1. 계량기 → 온수기(50gal) → 욕실 → 기구
2. 관경: 3/4in 주관, 1/2in 분기, 3/8in 세면기
3. 온수 대기 배관 최소화 (에너지)
4. 진공방지기·압력계 옥내 설치 검토

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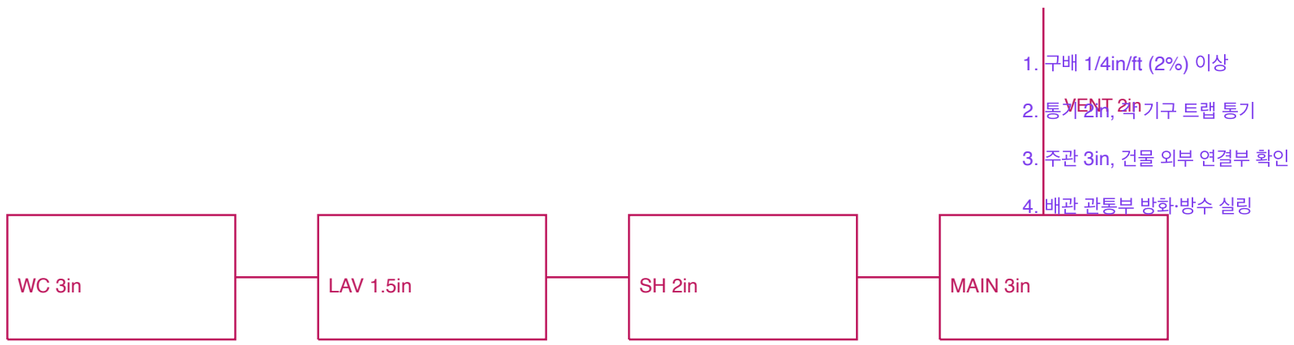
AFH 전환 (기존 3-car garage) - 허가 제출용 스키매틱

원안: Canarch Designs (Caleb Mongare) - A01.3 / A01.4

Project No. CMN/25/122 Date OCT 2026

SHEET P-03 급수 계통도 (개략)

이 도면은 A01.3/A01.4 기반 스키매틱입니다.
시공 전 실측 확인 필요. owner-builder 제출, 서명 불요.



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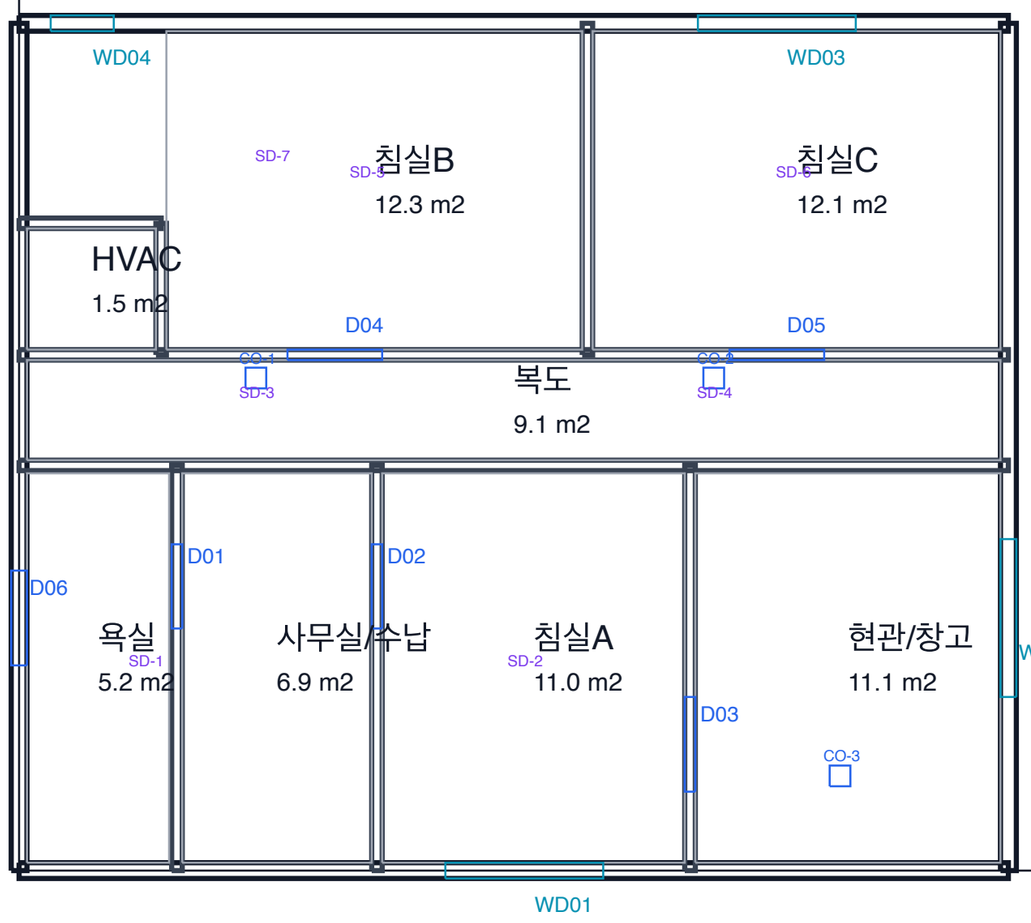
AFH 전환 (기존 3-car garage) - 허가 제출용 스키매틱

원안: Canarch Designs (Caleb Mongare) - A01.3 / A01.4

Project No. CMN/25/122 Date OCT 2026

SHEET P-04 배수·통기 계통도 (개략)

이 도면은 A01.3/A01.4 기반 스키매틱입니다.
시공 전 실측 확인 필요. owner-builder 제출, 서명 불요.



SD = 연기감지기 / CO = 일산화탄소 감지기 (전부 상호연동)

SD: 침실 내부 + 침실 출입구 인접 + 복도 (IRC R314.3)

CO: 취침구역 인접 + 차고 인접 구역 (IRC R315.5)

전원: 상시전원 + 배터리 백업(정전 7일), WAC R314.4

감지기: 조명/팬에서 3ft 이격, 벽-천장 모서리 12in 이내

이그레스 창: 침실A/B/C 3개소 (순개구부 5.7 sqft 이상 확인)

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AFH 전환 (기존 3-car garage) - 허가 제출용 스키매틱

원안: Canarch Designs (Caleb Mongare) - A01.3 / A01.4

Project No. CMN/25/122 Date OCT 2026

SHEET LS-01 연기·CO 경보 배치도

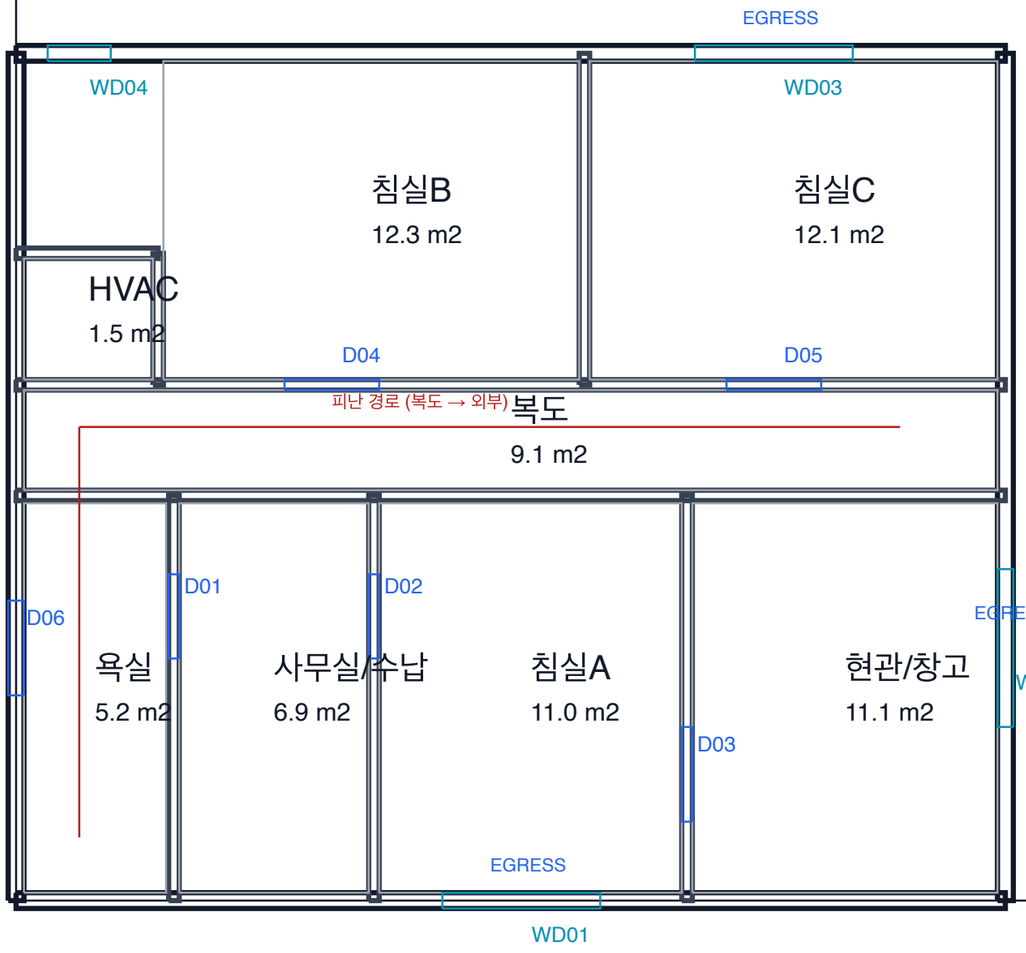
이 도면은 A01.3/A01.4 기반 스키매틱입니다.
시공 전 실측 확인 필요. owner-builder 제출, 서명 불요.

1. 침실 3개소 이그레스 창 (순개구부 5.7 sqft, IRC R310)

2. 복도 유효폭 36in 이상 유지

3. 문 개구부 32in 순폭 (R311.2)

4. 경보는 LS-01 참조



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AFH 전환 (기존 3-car garage) - 허가 제출용 스키매틱

원안: Canarch Designs (Caleb Mongare) - A01.3 / A01.4

Project No. CMN/25/122 Date OCT 2026

SHEET LS-02 이그레스·피난 경로

이 도면은 A01.3/A01.4 기반 스키매틱입니다.
시공 전 실측 확인 필요. owner-builder 제출, 서명 불요.

부위	면적(m2)	면적(sf)	가정 U	비고
외벽(순)	95.9	1032	0.048	R-21
천장	78.1	841	0.020	R-49
바닥	78.1	841	0.055	슬래브
창호	5.9	64	0.30	U-0.30
문	10.7	116	0.20	내부문

UA 합계 ≈ 14.4 W/K (예비 개산, 침투·열교 제외)

1) WSEC 2021(워싱턴) 주거 외피·창호·기밀 기준 충족 확인 필요

2) 창호 U-0.30 이하, SHGC 지역 상한 확인 (Zone 5 marine)

3) 외벽 R-21 + 외장 단열 또는 R-13+ci 대안 검토

4) 천장 R-49, 슬래브 주변 단열, 관통부 기밀(블로우도어 테스트)

5) 정식 컴플라이언스는 WSEC 툴/에너지 분석가 검토로 확정

6) 환기: 욕실 80 CFM, 주방 후드, 전체 환기 1대 (VIAQ)

이 표는 예비 개산이며 WSEC 컴플라이언스 문서가 아닙니다.

3571 Cassie Drive SW I Tumwater, WA 98512

LOT 7, CHAPPAREL LANDING DIVISION 2 · Tax-ID 4038-00-00700

AFH 전환 (기존 3-car garage) - 허가 제출용 스키매틱

원안: Canarch Designs (Caleb Mongare) - A01.3 / A01.4

Project No. CMN/25/122 Date OCT 2026

SHEET EN-01 WSEC 외피 요약 (예비)

이 도면은 A01.3/A01.4 기반 스키매틱입니다.
시공 전 실측 확인 필요. owner-builder 제출, 서명 불요.

MARK	크기	U	SHGC	VT	비고
WD01	5'-0"x4'-0"	0.30	0.25	0.45	IG/Argon
WD02	5'-0"x4'-0"	0.30	0.25	0.45	IG/Argon
WD03	5'-0"x4'-0"	0.30	0.25	0.45	IG/Argon
WD04	2'-0"x2'-0"	0.30	0.25	0.45	IG/Argon

1. WSEC 2021 주거 창호: U-0.30 이하, SHGC 상한 확인

2. 이그레스 창은 개폐 가능 타입 (slider/casement)

3. 창호 성능값은 제조사 인증서로 확인 필요

4. 설치 상세: 플래싱·에어실링 — A-15 참조

3571 Cassie Drive SW I Tumwater, WA 98512

LOT 7, CHAPPAREL LANDING DIVISION 2 · Tax-ID 4038-00-00700

AFH 전환 (기존 3-car garage) - 허가 제출용 스키매틱

원안: Canarch Designs (Caleb Mongare) - A01.3 / A01.4

Project No. CMN/25/122 Date OCT 2026

SHEET EN-02 창호 에너지 성능표

이 도면은 A01.3/A01.4 기반 스키매틱입니다.
시공 전 실측 확인 필요. owner-builder 제출, 서명 불요.

구성	면적(SF)	목표 R/U	비고
외벽 (순)	1032	R-21 + ci	2x6 + 외장단열
천장	841	R-49	기존 지붕 장선
바닥(슬래브)	841	R-10 주변	기존 슬래브
창호	64	U-0.30	IG/Argon
문	116	U-0.20	내부문 위주

1. WSEC 2021 주거용: 위 조합은 프리스크립티브 경로의 '목표' 값입니다.

2. 정식 컴플라이언스는 WSEC 컴플라이언스 툴 또는 에너지 분석가 계산서로 제출해야 합니다.

3. 기밀: 블로우도어 테스트 결과(ACH50)를 시트에 기재해야 합니다.

4. 환기(VIAQ): 욕실 80 CFM / 전체 환기 1대 — M-02, M-03 참조.

5. 이 표는 계산서가 아닌 요약이며, 수치 확정은 분석가 업무입니다.

3571 Cassie Drive SW | Tumwater, WA 98512

LOT 7, CHAPPAREL LANDING DIVISION 2 · Tax-ID 4038-00-00700

AFH 전환 (기존 3-car garage) - 허가 제출용 스키매틱

원안: Canarch Designs (Caleb Mongare) - A01.3 / A01.4

Project No. CMN/25/122 Date OCT 2026

SHEET EN-03 WSEC 외피 성능 요약 (제출용 표)

이 도면은 A01.3/A01.4 기반 스키매틱입니다.
시공 전 실측 확인 필요. owner-builder 제출, 서명 불요.

3571 Cassie Drive SW — Submission Package & Appendix Index

Tumwater, WA 98512 · Tax-ID 4038-00-00700 · Lot 7, Chapparel Landing Div. 2 · Owner-builder submittal · Prepared Oct 24, 2026

1. Document Register — main set (43 sheets)

Block	Contents	Sheets	Purpose
G-00 – G-06	Cover, general notes, legend, property data, permit application data, submittal checklist, discrepancy log	7	Administrative and code basis
C-01	Site & parking plan (1:200)	1	Lot 9,357 SF survey basis, 37.6 % coverage, 3 parking
A-01 – A-18	Existing / proposed plans, dimension, finish, RCP, furniture, schedules, section, details	18	Architectural scope of work
S-01 – S-03	Structural plan, foundation/slab, roof framing	3	Existing structure retained; no new loads
M-01 – M-03	Mechanical concept, equipment plan, duct system	3	EF-1 / AHU / CU-1 / DH-1
E-01 – E-03	Electrical plan, lighting, power riser	3	Circuits C1–C7
P-02 – P-04	Plumbing plan, supply diagram, drain/vent diagram	3	PEX / DWV
LS-01 – LS-02	Smoke & CO plan, egress plan	2	IRC R314 / R315, R310
EN-01 – EN-03	Envelope summary, window performance, WSEC summary	3	WSEC 2021 basis

2. Appendix Register — construction preparation

App.	Title	Source documents	Pages	Use
A	Project Inputs & Evidence	Owner sketches (before/after), recorded survey L2-25T2-3R, county assessor characteristics, county tax/legal record	5	Traceability: what the design was generated from
B	Construction Specification & Procedures	Master procedure/specs table (13 phases), master specs appendix, Review 2 QA markups	15	How the work is built; open review items
C	Materials, Budget & Cash Flow	Combined master booklet v3, master materials/specs/inspection booklet, budget reconciliation	4	Cost basis, VE strategy, draw schedule
D	Site QA & Delivery Control	Jobsite inspection posting card (inspection matrix), material delivery receipt log	2	Site posting and material intake control

Duplicate handling: the two uploaded “3571_construction_specs” files are byte-identical (SHA-256 match) — only one is bound in Appendix B. The one-page “Combined Master Booklet (V3)” and the 3-page “Master Materials, Specs & Inspection Booklet” overlap in content but are not identical; both are retained in Appendix C, booklet first, as summary then detail.

3. Budget Reconciliation — items requiring the owner's decision

The four points below are inconsistencies found while assembling the appendix. They are recorded, not resolved, and must be settled before the numbers are relied on by the City or an investor.

R-1 · Material takeoff exceeds the stated ceiling.
Every specification PDF in the appendix quotes a value-engineered material total of **$27,711.09** and the workbook row is labelled "TARGET ACHIEVED: Total Budget ≤ $30,000". Recomputing the 112 priced rows of the takeoff (qty × unit price) gives **$28,962.50**, and adding the workbook's own 15 % contingency gives **$33,306.88**. That is **$1,251.41 above** the quoted figure and **$3,306.88 above** the $30,000 ceiling.
Action: re-price or re-scope items 1–112, or raise the stated ceiling. Until then the appendix states both figures.

R-2 · Permit valuation vs. contract values.
The jobsite posting card states a permit valuation of **$52,711.09**. The construction documents carry a GC base labour contract of **$21,500.00** (10 % retainage = $2,150.00) and a material budget of roughly $27.7k–$33.3k, i.e. about **$49.2k–$54.8k**. The three figures are not derived from one another.
Action: confirm the valuation to be declared on the city application.

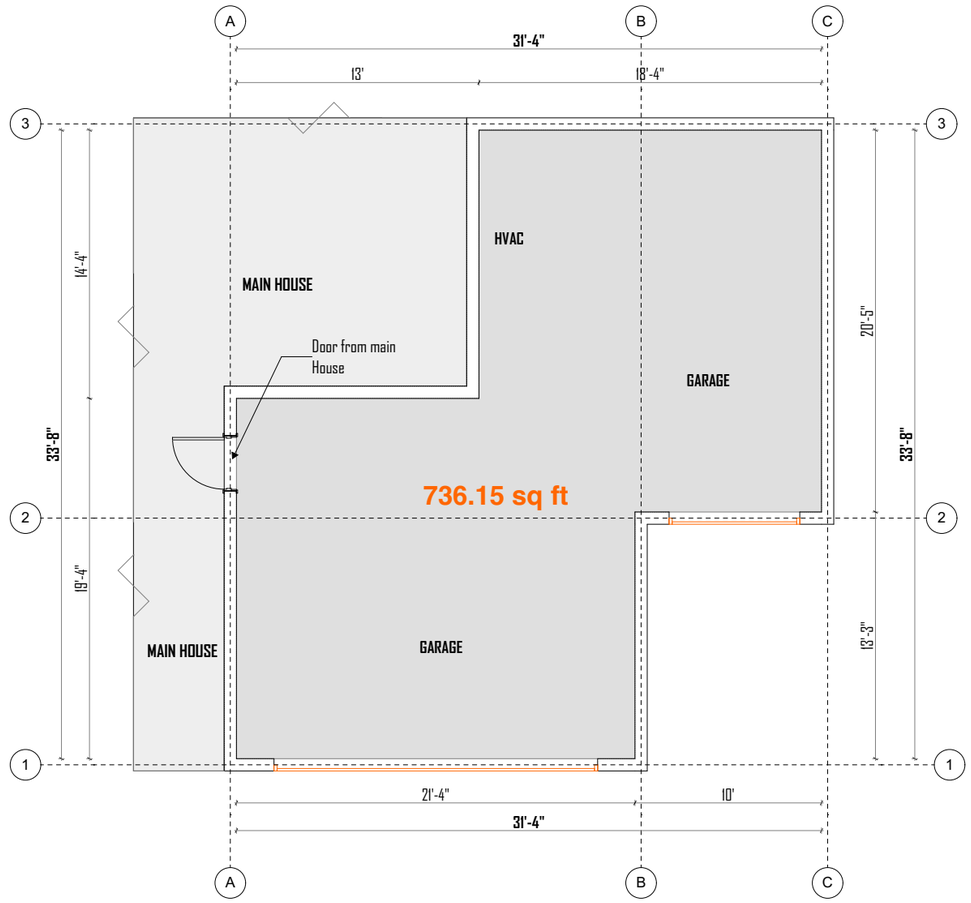
R-3 · Two different floor-area figures for the same conversion.
The specification set describes a **738 SF** conversion; the county record shows a **790 SF** attached garage and the architect's A01.3/A01.4 state **790 SF existing / 736 SF internal**. Sheet A-12 uses the A01.4 basis.
Action: declare one area on the application (recommend A01.4: 790 SF existing, 736 SF internal).

R-4 · Lot identity and area.
County legal description: **Lot 7**, 9,411 SF. Recorded survey cover sheet: **Lot 1**, 9,357 SF. MLS: 9,408 SF. Sheet G-06 carries the conflict; C-01 uses the survey area for coverage (37.6 %).
Action: confirm the lot number on the plat before uploading the survey as an appendix.

4. What is complete and what is not

Item	Status	Note
43-sheet drawing set (A3)	Complete	Owner-builder wording, no stamp required per city staff
Appendix A–D	Complete	Duplicates removed, order logical, indexed above
Permit application form (city PDF)	Not included	Download from ci.tumwater.wa.us; G-04 supplies the field values
Applicant / owner signature block	Blank by design	Name, contact, owner-builder declaration
Gas water heater relocation plan	Flagged	Unit currently inside garage (MLS); IRC M1307 / G2408 review
WSEC compliance calculation	Summary only	EN-01/EN-03 give targets; formal calculation is a separate sheet
Budget figures	Open	See R-1, R-2 above

This index is generated from the package itself; page counts in the Appendix register were verified when the appendix PDFs were bound.



GROUND FLOOR (Existing)

L{01}01

SCALE: 1/4" = 1'-0"

Total internal floor area=736 sq ft

CONSTRUCTION

NOTES: 1. GENERAL 1.1 Copyright of this drawing & all documentation pertaining to the project issued by the Architect is reserved by the Architect 1.2 This drawing is not to be scaled. Use written dimensions as indicated only 1.3 It is the responsibility of the contractor to check all setting out dimensions before commencing any work. Any discrepencies to be reported to the Architect immediately. This includes levels, heights, boundaries & benchmarks. 1.4 All floor levels indicated on the drawing are unfinished levels.

CANARCH DESIGNS
Architecture & construction

Canarch Designs
Cell no : 0727 726 819
Email : calebmongare@gmail.com

Drawing description

PROPOSED RESIDENTIAL ALTERATIONS – CONVERSION OF EXISTING GARAGE INTO LIVING SPACE, LOT 7, CHAPPAREL LANDING DIVISION 2

Scheme Design

Ground Floor Plan -Before

Revisions

Existing Use:
3-Car Garage

Proposed Use:
Residential / AFH

Proposed Bedrooms:
3 Bedrooms

Additional Space:
1 Office / Storage
1 Integrated Shower, Toilet & Sink

Ceiling Height:
10'-5"

Existing Garage Area: 790 SF

Internal floor area: 736 SF

Owner/Client/Agent

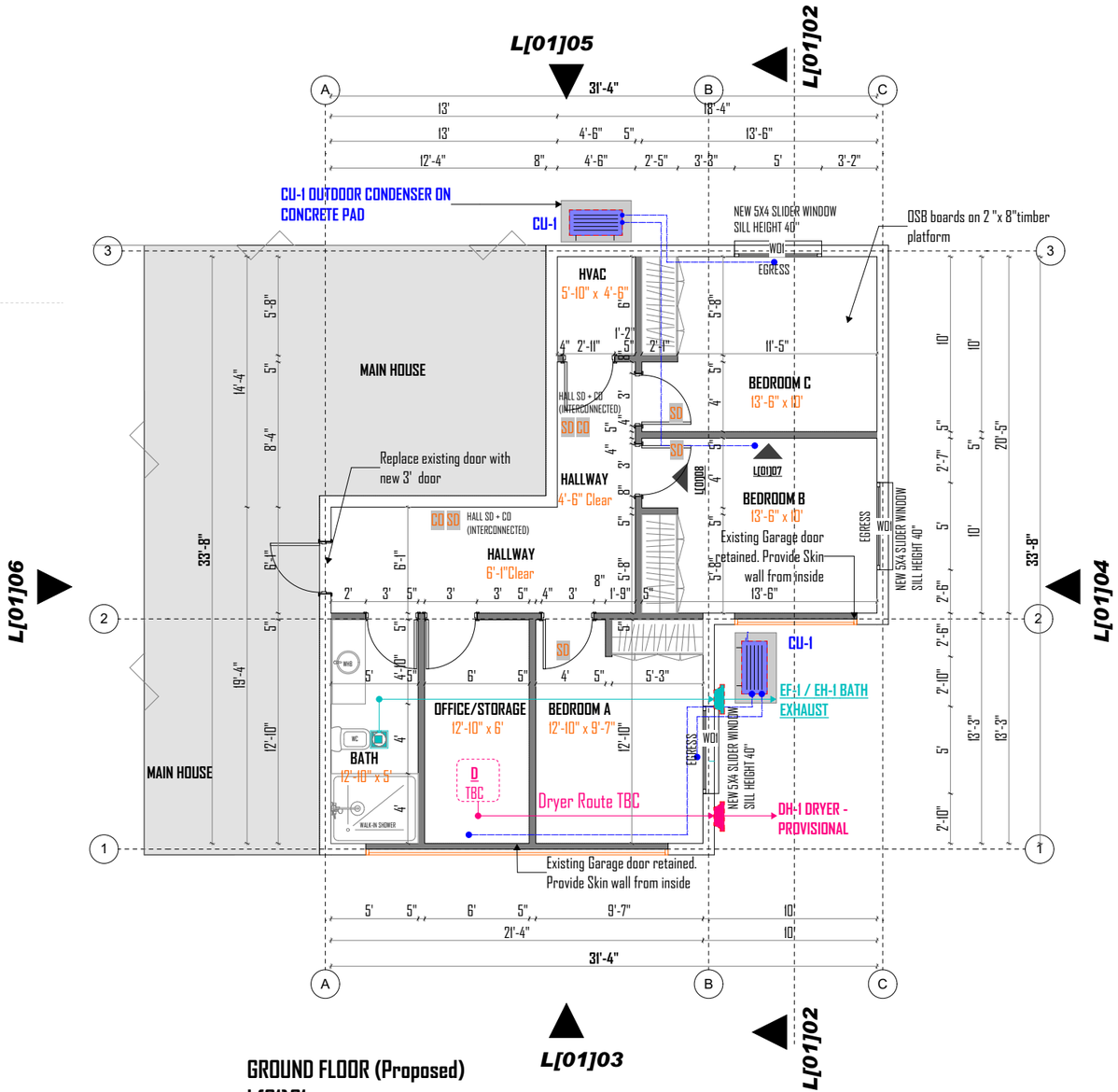
3571 Cassie Drive SW
Tumwater, WA 98512

Architect

Registration no.

Drawn by	CMN/25/122
Scale	1:50
Project no.	
Date	OCT 2026
Drg. no.	Rev.

A01.3



GROUND FLOOR (Proposed)
L{01}01
SCALE: 1/4" = 1'-0"
Total internal floor area=736 sq ft

LEGEND

- Existing wall
- New Partition wall
- Existing Door
- New Door
- New Window
- SD Smoke Detector
- CO Carbon monoxide Detector

SMOKE DETECTOR & CARBON MONOXIDE DETECTORS MUST BE INTERCONNECTED WITH BATTERY BACK UP PER WAC R314.4 AND IRC 314.4, R315.5

MECHANICAL ROUTES
PRELIMINARY / VERIFY

Proposed overlay on supplied floor plan | Schematic - do not scale

KEYED COORDINATION NOTES

CU-1 OUTDOOR CONDENSER

Refer to plan.
Concrete pad size, wall offset, airflow and service access to suit the selected unit.
Final position depends on verified site / HOA restrictions.

----- REFRIGERANT ROUTES

Sleeve / seal penetration. Indoor unit(s), final connection and pipe sizes are NOT confirmed.

EF-1 / EH-1 BATH EXHAUST

Proposed ceiling fan and independent overhead duct to bottom side exterior wall.
Provide exterior hood with backdraft damper. Confirm duct size and hood height.

D/ DH-1 DRYER - CONDITIONAL

D is only a placeholder in Office/Storage. Dryer location to be confirmed before using this route.
Route above ceiling to exterior side wall; verify chase / framing.
Separate duct; backdraft damper; NO terminal screen. [1]

CONSTRUCTION

CANARCH DESIGNS
Architecture & construction

Canarch Designs
Cell no : 0727 726 819
Email : calebmongare@gmail.com

Drawing description
PROPOSED RESIDENTIAL ALTERATIONS – CONVERSION OF EXISTING GARAGE INTO LIVING SPACE, LOT 7, CHAPPAREL LANDING DIVISION 2
Scheme Design

Ground Floor Plan- New

Revisions

Existing Use:
3-Car Garage

Proposed Use:
Residential / AFH

Proposed Bedrooms:
3 Bedrooms

Additional Space:
1 Office / Storage
1 Integrated Shower, Toilet & Sink

Ceiling Height:
10'-5"

Existing Garage Area: 790 SF

Internal floor area: 736 SF

Owner/Client/Agent
3571 Cassie Drive SW
Tumwater, WA 98512

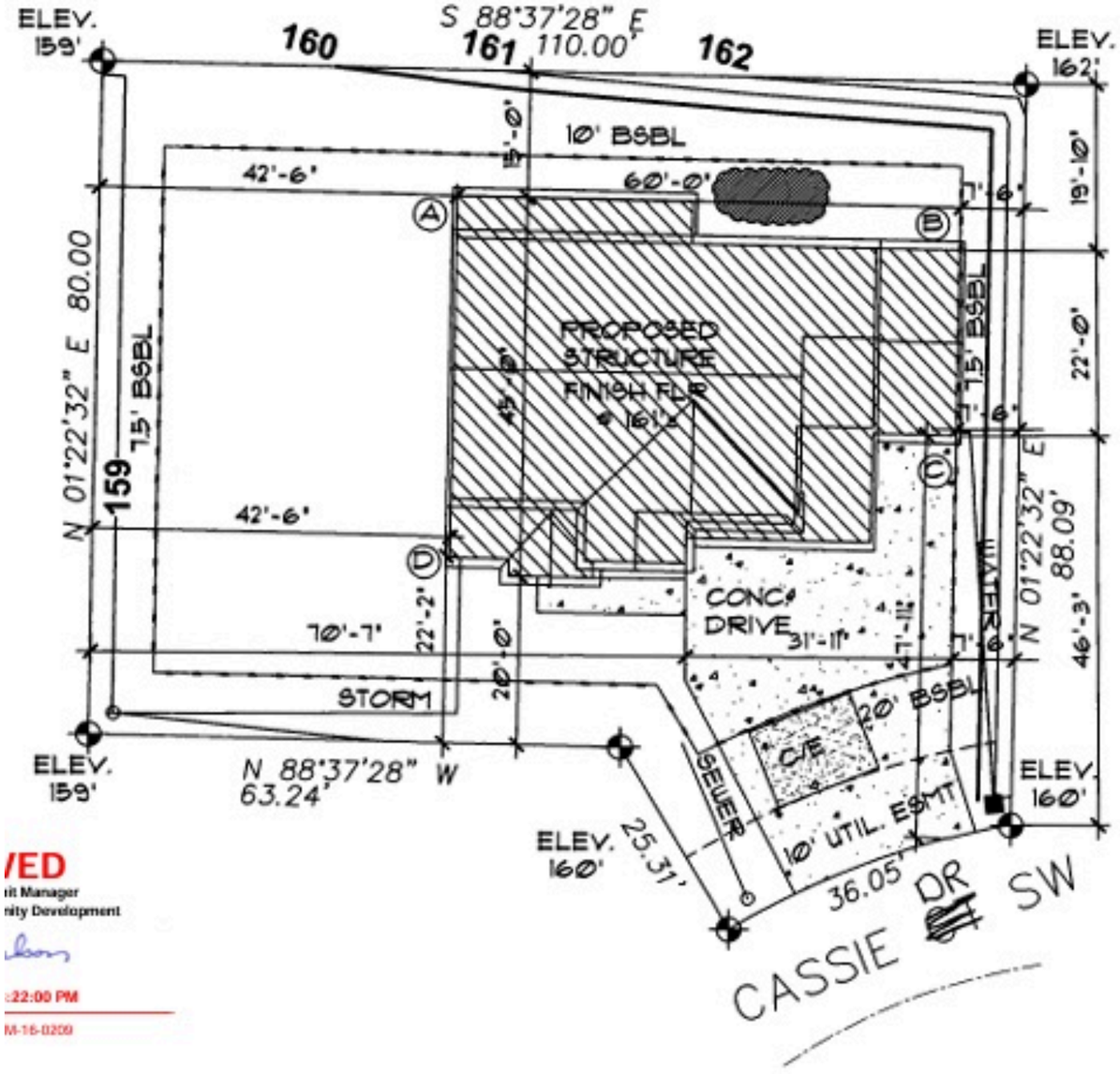
Architect

Registration no.

Drawn by	CMN/25/122
Scale	1:50
Project no.	
Date	OCT 2026
Drg. no.	Rev.

A01.4

NOTES: 1. GENERAL 1.1 Copyright of this drawing & all documentation pertaining to the project issued by the Architect is reserved by the Architect 1.2 This drawing is not to be scaled. Use written dimensions as indicated only 1.3 It is the responsibility of the contractor to check all setting out dimensions before commencing any work. Any discrepencies to be reported to the Architect immediately. This includes levels, heights, boundaries & benchmarks. 1.4 All floor levels indicated on the drawing are unfinished levels.



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CITY OF TUMWATER
COMMUNITY DEVELOPMENT

CHAPPAREL LANDING DIV. II

LOT 7
TUMWATER, WA
ZONING • R/SR

COVERAGE CALC.

LOT AREA•	9,397 SF
PROPOSED STRUCTURE•	2,214 SF
DRIVE/WALKWAY•	1,316 SF
LOT COVERAGE•	37.6 %

SCALE 1:20

OWNER INFO:

J. KERZIE HOMES, LLC.
253-377-6400

PLAN L2-2572-3R

7/30/15

Characteristics

Lot Acres	0.216	Fireplaces	1
Lot Area	9,411	Condition	Average
Land Use	SFR	Quality	Good
County Land Use	Residential Single	Water Source	Public
State Land Use	Residences	Sewer	Public Service
Year Built	2016	Heat Type	Heat Pump
Effective Year Built	2016	Porch	Covered Porch
Total Living Sq Ft	2,621	Porch Type	Covered Porch
Above Gnd Sq Ft	2,621	Patio Type	Patio
Gross Area	3,411	Patio/Deck 1 Area	240
Ground Floor Sq Ft	1,160	Garage Type	Built-In
Stories	2	Parking Type	Built-In Garage
Bedrooms	4	Attached Garage SF	790
Total Baths	Tax: 3 MLS: 2.5	Roof Type	Gable
MLS Total Baths	2.5	Roof Material	Composition Shingle
Full Baths	2	Roof Shape	Gable
Half Baths	1	Exterior	Wood Siding
Bath Fixtures	14	Foundation	Concrete

Tax

Tax Information

Tax-ID	4038-00-00700
Parcel ID	40380000700
Legal Description	SECTION 28 TOWNSHIP 18 RANGE 2W QUARTER NE SE & SE SE PLAT CHAPPAREL LANDING DIVISION 2 LT 7 DOCUMENT 4490930

GARAGE-TO-AFH CONVERSION — APPENDIX: MASTER CONSTRUCTION PROCEDURE & SPECS TABLE

Property: 3571 Cassie Dr SW, Tumwater, WA 98512 | Scheme Design: 16" Raised Insulated Floor Platform | Target Ceiling: ≤ $30,000 Total

Overview & Purpose: This master appendix provides a physical construction specification matrix for converting the 738 SF attached garage at 3571 Cassie Drive SW into a 3-bedroom Adult Family Home (AFH) suite. Based on the 16-inch raised insulated sleeper platform design (Submittal 05102026 / Canarch Designs), this table defines the exact sequence, predecessor/successor dependencies, tool requirements, trade man-hours, worksite locations, itemized material requirements, and step-by-step physical execution procedures for all 13 construction phases.

Step / Phase	Category & Location	Materials Needed	Tools & Equipment	Man-Hrs / Crew	Timeline & Sequence	Step-by-Step Construction Procedure & Technical Specs
1 Step 1: Site Prep & Safety Containment	Site Prep & Safety **Location:** Main Garage & House Threshold	• 6-mil Poly Sheeting (2 rolls) • Dust Barrier Zipper Kits (2) • Heavy Ram Board (2 rolls)	Utility knives, staple gun, masking tape, heavy broom, shop vac, trash bins	**12 Hrs** (2 Workers x 6 Hrs)	**Week 1, Days 1-2** **Prior:** HOA & City Permit Approval **Post:** Subfloor & Wall Framing	**1. Demolition & Clear-Out:** Remove existing garage door tracks, openers, wall shelving, and clear all debris from slab. **2. Dust Isolation:** Seal opening to existing residence with 6-mil flame-retardant poly sheeting and double zipper door kit. **3. Slab Inspection & Prep:** Scrap slab clean, check for high spots or major cracks, vacuum entire 738 SF concrete surface. **4. Surface Protection:** Lay Ram Board along path of travel into main residence.
2 Step 2: 16" Raised Floor & Partition Framing	Framing & Subfloor **Location:** 738 SF Garage Slab Footprint	• 2x12x20 PT Sleepers (38 pcs) • 3/4" T&G; OSB (28 sheets) • R-30 Subfloor Batts (14 bags) • Joist Hangers & Screws (1 lot) • 2x4x10 KD Studs (132 pcs) • PT 2x4 Bottom Plates (12 pcs) • Sill Seal & Anchors	Pneumatic framing nailer, circular saw, laser level, impact driver, rotary hammer drill, chalk line	**48 Hrs** (3 Workers x 16 Hrs)	**Week 1, Days 3-5** **Prior:** Step 1 Site Prep **Post:** Shell, Windows & Doors	**1. Capillary Break:** Lay 6-mil poly vapor barrier and closed-cell foam sill seal gasket directly over garage slab. **2. Raised Floor Platform:** Anchor PT 2x12 floor sleepers @ 12" O.C. using 1/2" x 3-3/4" wedge anchors to level with main house floor (16" rise). **3. Platform Insulation & Subfloor:** Friction-fit R-30 insulation batts between joist cavities; glue & screw 3/4" T&G; OSB subfloor panels. **4. Garage Skin Wall:** Frame 2x6 insulated skin wall directly behind retained overhead garage doors. **5. Interior Partitions:** Snap wall lines for Bedrooms A, B, C, Bath, Office & Mech room; erect 2x4 walls @ 16" O.C. (10'-5" ceiling height) w/ double top plates and 2x8 solid wood blocking for bathroom grab bars.
3 Step 3: Shell Sheathing, Windows & Egress Door	Exterior Envelope **Location:** Side (West/North) & Rear Walls	• 1/2" CDX Plywood (8 sheets) • 3 Vinyl Egress Windows (5.7 SF clear opening) • 1 HVAC Window • 1 ADA Exterior Door (36") • Flashing Tape & Sill Pans	Reciprocating saw, circular saw, level, caulk gun, hammer, scaffolding/stepladders	**24 Hrs** (2 Workers x 12 Hrs)	**Week 2, Days 6-8** **Prior:** Step 2 Wall Framing **Post:** MEP Rough-In	**1. Structural Sheathing:** Fasten 1/2" CDX plywood over exterior infill framing cavities. **2. Window Rough Openings:** Frame LVL/2x8 structural headers and jack studs for 3 egress windows on side/rear elevations and 1 HVAC utility window. **3. Egress Window Setting:** Install flexible sill flashing; set vinyl slider egress windows plumb & level; verify sill height ≤ 44" AFF and clear net opening ≥ 5.7 sq ft per IRC R310. **4. Exterior Door:** Set 36" insulated exterior fiberglass door with low-profile ADA sill pan and polyurethane weather sealant.

Step / Phase	Category & Location	Materials Needed	Tools & Equipment	Man-Hrs / Crew	Timeline & Sequence	Step-by-Step Construction Procedure & Technical Specs
4 Step 4: MEP Rough-In (Plumbing & Electrical)	Mechanical Trades **Location:** Wall Cavities, Attic & Bath Slab	• PEX-A 1/2" & 3/4" Tubing • DWV PVC Pipe 2" & 3" • ASSE 1070 Mixing Valve • Washing Machine Box • 14/2, 12/2, 14/3 Romex Wire • Vapor-Tight Outlet Boxes • AFCI/GFCI Breakers	PEX expansion tool, PVC saw & cement, wire stripper, drill w/ auger bits, voltage tester, fish tape	**40 Hrs** (2 Plumbers + 2 Electricians x 10 Hrs)	**Week 3, Days 11-15** **Prior:** Step 3 Shell Complete **Post:** City Rough Inspection & Insulation	**1. Plumbing Supply & Drain:** Core slab or route 3" DWV sewer line to main house stack; run 2" branch lines for curbless shower & laundry; run 3/4" PEX-A main supply & 1/2" branch runs; rough-in ASSE 1070 anti-scald mixing valve (≤120°F). **2. Electrical Circuits:** Pull 14/2 NM-B for 15A bedroom circuits, 12/2 NM-B for 20A bath/laundry circuits, and 14/3 for interconnected smoke alarms & 3-way switches; set vapor-tight boxes. **3. Inspection Call:** Secure City of Tumwater Rough Plumbing & Electrical signoffs.
5 Step 5: Thermal Insulation, Air-Seal & Soundproofing	Thermal Envelope **Location:** Exterior Walls, Ceilings & Partitions	• Fireblock Foam Kits (1 bundle) • R-21 Wall Batts (5 bags) • R-15 Perimeter Batts (9 bags) • R-38 Ceiling Batts (17 bags) • Rockwool Safe'n'Sound (14 bags) • Smart Vapor Retarder (1 roll)	Insulation knife, staple gun, caulk gun, extension ladders, PPE respirators/goggles	**16 Hrs** (2 Workers x 8 Hrs)	**Week 4, Days 16-18** **Prior:** Step 4 Rough MEP Passed **Post:** City Insulation Inspection & Drywall	**1. Air Sealing:** Seal all wire/pipe penetrations and top/bottom plates with fireblock expanding foam. **2. Thermal Insulation:** Friction-fit R-21 Kraft batts in 2x6 infill wall, R-15 in perimeter 2x4 walls, and R-38 in ceiling cavity (WSEC Zone 4C). **3. Acoustic Insulation:** Install Rockwool Safe'n'Sound mineral wool batts in all interior resident bedroom partitions and bathroom walls for STC sound isolation. **4. Vapor Membrane:** Staple Class II smart vapor retarder over exterior wall framing. **5. Inspection:** Secure City of Tumwater Insulation Inspection signoff.
6 Step 6: Drywall Boarding, Taping & Fire Separation	Wall Finishing **Location:** All Interior Walls, Ceiling & Garage Firewall	• 1/2" Interior Drywall (90 sheets) • 5/8" Type X Firecode (26 sheets) • Greenboard 1/2" (8 sheets) • Joint Compound (10 pails) • Paper Tape & Corner Bead • Drywall Screws & Acoustic Sealant	Drywall screw gun, drywall lift, T-square, taping knives (4", 6", 10", 12"), mud pan, pole sander	**40 Hrs** (2 Drywallers x 20 Hrs)	**Week 5, Days 21-25** **Prior:** Step 5 Insulation Passed **Post:** Flooring & Tile Substrate	**1. Fire Separation Assembly:** Hang 5/8" Type X fire-rated drywall on the dwelling separation wall and ceiling per IRC R302.6; tape w/ firecode compound. **2. Moisture-Resistant Board:** Hang 1/2" greenboard in bathroom dry areas. **3. Standard Boarding:** Hang 1/2" drywall across all resident bedrooms, office, and hallway ceilings/walls. **4. Joint Finishing:** Apply paper tape and 3 coats of joint compound to Level 4 smooth finish; apply acoustic sealant at perimeter floor joints; sand smooth for paint.
7 Step 7: Subfloor Leveling & 20-mil LVP Flooring	Finishes — Flooring **Location:** 650 SF Living Area (Excl. Bath Tile)	• Self-Leveler Compound (10 bags) • Concrete Primer (2 gals) • 20-mil Commercial LVP (37 boxes) • 6-mil Vapor Underlayment (1 roll) • ADA Reducer Strips (6 pcs)	Mixing paddle & 5-gal bucket, gauge rake, floor roller, guillotine tile cutter, tapping block, pull bar	**20 Hrs** (2 Installers x 10 Hrs)	**Week 6, Days 26-27** **Prior:** Step 6 Drywall Sanded **Post:** Waterproofing & Tile	**1. Subfloor Prep:** Prime OSB subfloor and apply self-leveling compound at seam transitions for a dead-flat surface (1/8" in 10' tolerance). **2. Moisture Barrier:** Roll out 6-mil poly underlayment film with 6" overlapped taped seams. **3. LVP Installation:** Lay 20-mil commercial wear layer rigid core LVP with 1/4" wall expansion gaps; stagger end joints minimum 8". **4. Transitions:** Fasten flush ADA beveled reducer strips at doorway thresholds to prevent trip hazards.

Step / Phase	Category & Location	Materials Needed	Tools & Equipment	Man-Hrs / Crew	Timeline & Sequence	Step-by-Step Construction Procedure & Technical Specs
8 Step 8: Water proofing, Curbless Shower & Bath Tile	Finishes — Bathroom **Location:** 86 SF Resident Bathroom Suite	• 1/2" Cement Board (10 sheets) • Schluter-Kerdi Membrane System • Curbless Shower Pan Kit • Linear Drain Assembly • DCOF Tile & Subway Wall Tile • Polymer Thinset & Grout	Notched trowels (1/4", 3/8"), wet tile saw, tile nippers, rubber float, sponge, laser level	**32 Hrs** (2 Tile Setters x 16 Hrs)	**Week 6, Days 28-30** **Prior:** Step 7 Subfloor Leveling **Post:** Doors, Millwork & Fixtures	**1. Wall Backer:** Fasten 1/2" cement backer board to shower walls w/ alkali-resistant screws and mesh tape. **2. Curbless Pan & Membrane:** Install pre-sloped barrier-free curbless shower pan flush w/ raised floor deck; apply Schluter-Kerdi waterproof sheet membrane up walls to 72" AFF; flood test 24 hrs. **3. Tile Setting:** Set DCOF ≥ 0.42 slip-resistant mosaic floor tile in shower/bath floor; set white ceramic subway wall tile on shower walls using modified polymer thinset. **4. Grouting:** Apply stain-resistant polymer grout; caulk all wall/floor transitions w/ 100% silicone sealant.
9 Step 9: Prehung Doors, ADA Locksets & Trim	Carpentry Finish **Location:** All Door Openings & Baseboards	• 36" Solid Core Interior Doors (5) • 20-Min Fire Door w/ Closer (1) • Louvered HVAC Door (1) • Bi-Fold Closet Doors (3 sets) • Grade 2 ADA Lever Handles (7) • Single-Action Exit Lockset (1) • Primed MDF Base & Casing	Finish nailer (18-gauge), miter saw, level, wood shims, hole saw kit, screwdriver set	**24 Hrs** (2 Carpenters x 12 Hrs)	**Week 7, Days 31-33** **Prior:** Step 8 Bath Tile Set **Post:** MEP Trim Out & Paint	**1. Door Hanging:** Set 36" solid-core interior prehung doors plumb and square using wood shims; verify 32" net clear passage width when open at 90 degrees. **2. Fire Door:** Set 36" 20-min fire-rated door w/ self-closing hinges at house threshold. **3. ADA Hardware:** Install Commercial Grade 2 ADA lever handles; install single-action interconnected lever lockset on exterior exit door (unlatches in single motion without keys). **4. Millwork Trim:** Miter & nail primed MDF Craftsman baseboards (3-1/4") and door/window casing (2-1/4"); fill nail holes w/ spackle.
10 Step 10: Bathroom Vent, HVAC Duct & Mini-Split	HVAC Finish **Location:** Bath Ceiling, Attic & Bedroom Walls	• Quiet 80 CFM Fan (<0.8 Sone) • R-6 Flexible Duct & Vent Hood • Rigid Metal 4" Dryer Vent Kit • R-8 Supply Ducting (60 ft) • Supply Registers & Return Grille • Door Transfer Grilles (3) • 24k BTU Mini-Split System	Duct crimper, hole cutter, foil tape, vacuum pump, gauge manifold, drill, duct mastic	**20 Hrs** (1 HVAC Tech + 1 Helper x 10 Hrs)	**Week 7, Days 34-35** **Prior:** Step 9 Doors Hung **Post:** Electrical & Paint Finish	**1. Bath Exhaust:** Mount quiet 80 CFM continuous-run exhaust fan (<0.8 Sone) in bath ceiling; route 4" R-6 insulated duct to exterior wall cap w/ backdraft damper (ASHRAE 62.2). **2. Duct Distribution:** Run 6"/8" R-8 insulated supply ducts from central unit to ceiling boots in bedrooms & common room; set registers & central return grille. **3. Air Balancing:** Install door undercut transfer grilles on bedroom doors to prevent pressure isolation. **4. Heat Pump:** Mount 24k BTU multi-zone mini-split indoor air handlers in Bedrooms A, B, C; pull line sets, evacuate system, and commission condenser.

Step / Phase	Category & Location	Materials Needed	Tools & Equipment	Man-Hrs / Crew	Timeline & Sequence	Step-by-Step Construction Procedure & Technical Specs
11 Step 11: Electrical Trim, Lighting & Safety Alarms	Electrical Finish **Location:** Switches, Outlets, Fixtures & Ceilings	• Tamper-Resistant Outlets (18) • Self-Test GFCI Outlets (3) • Decora Rocker Switches (9) • 6" LED Wafer Downlights (10-pack) • Wet-Location Shower Light (1) • Hardwired Smoke Alarms (3) • Hardwired Combo Smoke/CO (3) • UL 924 Emergency Exit Light (1) • 5-lb ABC Fire Extinguishers (2)	Wire stripper, voltage tester, screwdrivers, wire nut driver, stepladder	**16 Hrs** (2 Electricians x 8 Hrs)	**Week 8, Days 36-37** **Prior:** Step 10 HVAC Installed **Post:** Painting & Final Inspection	**1. Outlets & Switches:** Wire tamper-resistant duplex outlets, GFCI outlets in bath/laundry, and Decora rocker switches; install polycarbonate wall plates. **2. LED Lighting:** Snap 6" canless ultra-thin LED wafer downlights into ceiling cutouts; install IP65 wet-location downlight over shower. **3. Life Safety Alarms:** Mount hardwired 120V interconnected smoke alarms inside Bedrooms A, B, C, and combo smoke/CO alarms in hallway, common room, and mech room. **4. Emergency Signage:** Mount UL 924 battery-backup emergency exit light over exit corridor; mount 2 5-lb ABC fire extinguishers.
12 Step 12: Wall Primer, Washable Paint & Accessories	Finishes — Paint **Location:** All Walls, Ceilings, Trim & Closets	• PVA Drywall Primer (3 pails) • Low-VOC Eggshell Paint (5 pails) • Semi-Gloss Trim Enamel (2 gals) • Caulking & Painting Supplies • Closet Organizers (3 sets) • ADA Bath Accessories (Mirror, Towel/Paper Dispensers, Shower Seat)	Paint roller frames, 3/8" nap covers, sash brushes, drop cloths, caulking gun, drill	**24 Hrs** (2 Painters x 12 Hrs)	**Week 8, Days 38-39** **Prior:** Step 11 Electrical Trim **Post:** City Final & DSHS License	**1. Surface Prep & Prime:** Caulk all trim/drywall seams; apply 1 coat of PVA drywall primer over fresh sheetrock. **2. Wall Painting:** Apply 2 coats of washable low-VOC eggshell interior latex paint in resident bedrooms, hallway, and common area. **3. Trim Enamel:** Apply 2 coats of semi-gloss white enamel on all interior doors, window casings, and baseboards. **4. Hardware & Storage:** Install ventilated double-bar closet shelf systems in resident wardrobes; securely anchor wall-hung tilt mirror, towel dispensers, paper holders, and phenolic folding shower seat (400-lb rating).
13 Step 13: GC On-Site Verification (HOLD Scopes)	Contractor Verification **Location:** Main Service Panel & Exterior Grade	• 200A/400A Main Service Panel Upgrade / Feeders (if needed) • Manual J Heat Pump Sizing Verification	Electrical multitester, clamp meter, HVAC airflow anemometer, civil laser level	**8 Hrs** (GC Lead x 8 Hrs)	**Week 8, Day 40** **Prior:** Step 12 Finishes Complete **Post:** City Occupancy Signoff & DSHS Licensing	**1. Electrical Load Calculation:** Electrician to verify load demand on existing 2016 200A home service panel before final signoff. **2. HVAC Balancing:** Verify multi-zone heat pump heating/cooling performance and temperature differential across all bedrooms. **3. City Final Inspection:** Accompany City of Tumwater Building Inspector for final Certificate of Occupancy signoff. **4. DSHS AFH Licensing Inspection:** Conduct walkthrough w/ WA State DSHS licensor to verify grab bar anchoring, anti-scald hot water temp (≤120°F), interconnected alarms, and emergency exit lighting.

MASTER CONSTRUCTION PROCEDURE & TECHNICAL SPECS APPENDIX

3571 Cassie Drive SW, Tumwater, WA 98512 | Parcel #40380000700 (Lot 7, Div 2) | 790 SF Footprint

Master Construction Specs Overview (790 SF Conversion):

- **Step 1 (Site Prep):** 6-mil poly containment, zipper doors, Ram Board floor protection over 790 SF slab.
- **Step 2 (16" Raised Floor):** 38 PT 2x12 sleepers @ 12" O.C., R-30 insulation, 3/4" T&G; OSB deck, 2x4 framing @ 16" O.C.
- **Step 3 (Shell & Windows):** 1/2" CDX plywood, 3 vinyl egress windows (sill ≤ 44" AFF, clear opening ≥ 5.7 SF), 1 utility window, 36" ADA door.
- **Step 4 (MEP Rough):** 3" DWV sewer & 2" drains, PEX-A supply, ASSE 1070 mixing valve (≤ 120°F), Romex wiring. City Line: (360) 754-4180.
- **Step 5 (Insulation):** R-21 wall, R-15 perimeter, R-38 ceiling batts, Rockwool Safe'n'Sound acoustic batts.
- **Step 6 (Drywall):** 5/8" Type X firecode drywall on garage separation wall/ceiling (IRC R302.6), Level 4 finish.
- **Step 7 (Waterproofing & Tile):** Pre-sloped curbless shower pan, Kerdi membrane (24-hr flood test), DCOF ≥ 0.42 floor tile.
- **Step 8 (Finishes & WABO Signoff):** 20-mil commercial LVP, 36" solid interior doors, ADA levers, Decora switches, LED wafer lights, smoke/CO alarms, UL 924 exit sign, 4 grab bars w/ 2x8 backing. Final inspection & **WABO Ready Signoff**.

A01.1 • Cover Sheet

Source: “3571 Cassie Drive SW Submittal 13102026.pdf”, page 1. Review 2 page 1 corresponds to the same source page.

Use printed Scope of Work / General Notes item numbers. Unnumbered title-block and footer notes are identified by exact wording.

ID / status	Exact point on the source page / current wording	Finding and requested correction / decision
1A Resolve	Upper-left Scope of Work, item 4: “CONSTRUCT A RAMP RISING 16″ FROM THE GRADE.”	Conflicts with 6″ on A01.8 (8A). Architect: verify final grade/floor rise and revise cover, site, section and ramp together. Do not choose a rise by assumption.
1B Clarify	Upper title: “FUTURE ADULT FAMILY HOME”; General Notes item 2: “2021 INTERNATIONAL BUILDING CODES”; right title block: “Scheme Design”; diagonal “CONSTRUCTION.”	State permit phase and one issue status. List applicable adopted codes with WA/Tumwater amendments. Homeowner: confirm conversion now versus complete AFH application. Add whole-home AFH plan rather than imply garage-only plans establish licensing compliance.
1C Add detail	Scope items 10–12: “INSULATE / FINISH PER ... WSEC”; “NEW PLUMBING”; “NEW ELECTRICAL.”	Provide WSEC/envelope assembly documentation and coordinated plumbing/electrical work, or identify separately approved documents. A generic compliance note is not the design. Heating/ventilation also need actual equipment/airflow information.
1D Rewrite	Lower-left General Notes items 3, 5 and 6: “RESPOSIBLE”; “REGULATIONOR”; “INTENTOF”; discrepancy/contractor responsibility wording.	Correct to RESPONSIBLE / REGULATION OR / INTENT OF. Suggested discrepancy note: “Verify field conditions. Report discrepancies to the design professional and obtain written clarification before proceeding with affected work.” Review liability/fine allocation separately.
1E Editorial	General Notes item 7: “(IRC 303.10)”; upper-right Index of Drawings begins A01.2; Scope item 5: “SEPARATING MECHANICAL...”	Use “IRC R303.10”; add Cover Sheet A01.1 to the index; rewrite item 5 as “ENCLOSE EXISTING HVAC EQUIPMENT IN A DEDICATED MECHANICAL CLOSET.”
1F Verify	Right title block: “Ceiling Height: 10′5″”, owner/address, blank Architect/registration/revision fields; bottom border note 1.4: “...unfinished levels.”	Correct height using 6A. Identify owner/preparer, full issue date and revision. Accessibility/opening heights must reference final finished surfaces, not blanket unfinished levels. Apply this correction to every sheet.

How to use: Drawings have no printed line numbers. Use sheet + note/view/grid + quoted text. Resolve = conflicting data; Verify = evidence needed; Add/Clarify = incomplete; Editorial = wording; Checked = preserve. IDs are review references, not original drawing labels.

Reply by ID: Item_____ □Agree □Clarify □Alternative Revised sheet/rev_____ Date_____
Architect response / homeowner decision:_____

Document review only, not construction authorization or engineering/licensing approval. Original Review and source drawings unchanged.

A01.2 • Site Plan

Source: “3571 Cassie Drive SW Submittal 13102026.pdf”, page 2. Review 2 page 2 corresponds to the same source page.

Positions below refer to the source drawing as displayed, not to geographic directions. Historical 2016 approval is reference only.

ID / status	Exact point on the source page / current wording	Finding and requested correction / decision
2A Verify	Lower-left legal/zoning block: “CHAPPAREL LANDING DIV. II / LOT 1”; right title description: “LOT 7”; cover parcel 40380000700.	Verify recorded legal description/parcel and use one correct lot number throughout. Do not infer the proper subdivision spelling or parcel boundary from this conflict.
2B Resolve	Upper-left red ramp note: “16′-0″ HORIZONTAL RUN ... 16″ VERTICAL RISE”; red strip dimensions 3′ / 16′ / 3′.	Coordinate with 1A, 6A and 8A. Specify verified finished endpoints, true sloped run excluding landings, clear rail-to-rail width, door operation and site connection. Apply both IRC and DSHS slope limits (8B).
2C Add plan	Red ramp at rear/patio; yellow garage-conversion footprint on right; whole main-house interior is not drawn.	Show complete resident route from each bedroom through permitted common circulation to designated exit/ramp and safe location. Include main-house/all-floor plan, thresholds and S/NS1/NS2 bedroom classifications. Do not assume an escape window is a second accessible exit.
2D Coordinate	Two blue MINISPLIT CONDENSOR symbols by right side of house; color legend beneath footprint.	Confirm one or two outdoor units; coordinate with two CU-1 plan symbols (4D). Give unique tags if two. Dimension selected-unit clearances, service access, egress conflicts, noise and HOA/site constraints. “CONDENSOR” → “CONDENSER.”
2E Checked	Lower-left COVERAGE CALC.: 9,397 SF lot; 2,214 SF structure; 1,316 SF drive/walkway; 37.6%. Corner block: total 638 / 4 = 159.5′.	Arithmetic checks: (2,214+1,316)/9,397 = 37.565%, rounded 37.6%; corner average 159.5′. Preserve unless inputs change. Explain whether new ramp/pads affect coverage or impervious-area calculations. Define footprint versus total floor area; do not equate 2,621 SF SFR with footprint.
2F Clarify	Bottom-center “SCALE 1:20”; title-block scales “1:100, 1:151.13, 1:143.33”; lower-middle 2016 approval stamp and reference label.	Use separately verified view scales or NTS/reference base. Keep “2016 SUBMITTAL - REFERENCE ONLY” prominent; it does not approve current work. Mark historical drainage/erosion details not in current scope where appropriate.

Reply by ID: Item_____ □Agree □Clarify □Alternative Revised sheet/rev_____ Date_____
Architect response / homeowner decision:_____

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A01.3 • Existing Ground Floor (Before)

Source: “3571 Cassie Drive SW Submittal 13102026.pdf”, page 3. Review 2 page 3 corresponds to the same source page.

This sheet provides existing control dimensions, but not a surveyed as-built condition or a whole-home AFH plan.

ID / status	Exact point on the source page / current wording	Finding and requested correction / decision
3A Checked	Plan perimeter, grid 1–3 / A–C: 13′3″+20′5″; 19′4″+14′4″; 21′4″+10′; 13′+18′4″.	These close to 33′8″ and 31′4″. Architect: field-verify the controls and adopt one canonical dimension set. Proposed plan, elevations and section must then use those same controls (4A, 5A, 6C).
3B Clarify	Orange center label “736.15 sq ft”; below-plan caption “Total internal floor area=736 sq ft”; right summary “Existing Garage Area:790 SF.”	736.15→736 is acceptable rounding. Define 790 gross-area boundary versus 736.15 internal footprint; difference is not itself an error. Relabel proposed usable room areas separately, after walls/closets/HVAC exclusions.
3C Add elevations	Left-middle callout “Door from main House”, grid A near 2; MAIN HOUSE labels at left; GARAGE area shaded gray.	Survey garage slab, main-house finished floor, connecting door threshold and exterior grade. Show these datums and demolition/retention extent. The future raised floor cannot be coordinated without the existing level relationship.
3D Add information	Upper central/right plan zone labeled HVAC; exterior garage openings at bottom and right edge near grid 2.	Document actual HVAC/appliance type, fuel, venting/service needs and opening framing. Identify existing utilities and any alterations. Do not presume a mechanical closet is adequate merely from the label.
3E Editorial	Below-plan view L{01}01: “SCALE:1/4″=1′-0″”; right title-block scale 1:50 and “Ground Floor Plan -Before.”	1/4″=1′ is 1:48, not 1:50. Verify plotted output and use one scale or “As shown.” Use “Existing Ground Floor Plan.” See page 10 for common title/footer changes.

Reply by ID: Item_____ □Agree □Clarify □Alternative Revised sheet/rev_____ Date_____
Architect response / homeowner decision:_____

Document review only, not construction authorization or engineering/licensing approval. Original Review and source drawings unchanged.

A01.4 • Proposed Ground Floor (After)

Source: “3571 Cassie Drive SW Submittal 13102026.pdf”, page 4. Review 2 page 4 corresponds to the same source page.

4A1–4A7 are written-dimension conflicts. Verify grid/face endpoints and canonical controls; regenerate all chains. Do not arbitrarily reduce a selected segment.

4A • Dimension discrepancies: locate by row, not by scaling the drawing

ID	Source point	Printed chain	Computed / control
4A1	Above plan, second A–C dimension row	13′+4′6″+5″+13′6″	31′5″ vs 31′4″
4A2	Below plan, innermost A–B row	5′+5″+6′+5″+9′7″	21′5″ vs 21′4″
4A3	Left detailed vertical row beside main-house outline,1–3	12′10″+5″+6′1″+8′4″+5″+5′8″	33′9″ vs 33′8″
4A4	Inside plan, vertical hall/Bedroom B–C partition,2–3	5′8″+5″+4′+5″+4′+5″+5′8″	20′7″ vs 20′5″
4A5	Inside plan, horizontal chain immediately above Bedroom A	4′+5″+5′3″	9′8″ vs 9′7″
4A6	Right detailed exterior/window row beside Bedroom B above grid 2	2′6″+5′+2′7″	10′1″ vs 10′
4A7	Right detailed exterior/window row, lower 1–2 portion	2′10″+5′+2′10″+2′6″	13′2″ vs 13′3″

Other coordinated corrections

ID / status	Exact point on the source page / current wording	Finding and requested correction / decision
4B Verify	Bedrooms A/B/C and wardrobes; HVAC in upper-center labeled 5′10″×4′6″; BATH bottom-left 12′10″×5′.	Add occupancy/net-usable-area and dimensioned bathroom schedules. A ≈ 122.99 SF before exclusions; B/C135 SF before exclusions; C minus HVAC 26.25 SF gives ≤ 108.75 SF before wardrobe. No demonstrated double occupancy for C; A has only 2.99 SF margin. Exclude closets/wardrobes/vestibules; 80 SF single/120 double. Show shower final clearance, bars/backing, fixtures/door circulation.
4C Add schedule	Middle-left leader “Replace existing door with new 3′ door”; MAIN HOUSE connections; window labels “SILL HEIGHT 40″”/EGRESS.	Nominal 3′ is not installed clear width. Schedule doors/thresholds and complete exit route/classifications (page 10). Show actual clear egress-window data (7B). Coordinate raised floor/retained-door wall with 6A/6B/6D. Homeowner: confirm intended single/double occupancy.
4D Finalize	Right MECHANICAL ROUTES / PRELIMINARY / VERIFY; two blue CU-1 symbols; conditional D/DH-1 dryer in Office/Storage.	Confirm one/two units and unique tags; specify indoor units, capacity, airflow, duct/pipe sizes, condensate/service/penetrations and actual termination façade. Homeowner: decide dryer inclusion/location. Remove placeholder/TBC/[1] or supply final design/source.
4E Rewrite	Right, directly below LEGEND: “...BATTERY BACK UP PER WAC R314.4 AND IRC 314.4, R315.5.”	Use listed smoke/CO “alarms”; correct complete WAC/IRC references, power/backup and whole-home interconnection/audibility. R314.4 alone is not the backup-power rule. Applicable alteration exceptions and visual alarms depend on current rules/resident needs (page 10).

Reply by ID: Item_____ □Agree □Clarify □Alternative Revised sheet/rev_____ Date_____
Architect response / homeowner decision:_____

A01.5 • Elevations

Source: “3571 Cassie Drive SW Submittal 13102026.pdf”, page 5. Review 2 page 5 corresponds to the same source page.

Four elevation views: L{01}03 upper-left,04 upper-right,05 lower-left,06 lower-right. Do not infer compass directions.

ID / status	Exact point on the source page / current wording	Finding and requested correction / decision
5A Resolve	Four views: upper-left L{01}03 A–B and lower-left L{01}05 B–A both 22′; upper-right L{01}04 2–3 and lower-right L{01}06 3–2 both 21′1″.	The 22′ dimensions conflict with plan 21′4″; the 21′1″ dimensions conflict with plan 20′5″: each differs by 8″. Verify controls then correct all related sheets. If plan controls are confirmed, use 20′5″/21′4″; these are not independently verified field values. Elevation totals otherwise imply 34′4″/32′ versus plan 33′8″/31′4″.
5B Verify	Two L{01}04 windows and one L{01}05 window: “New Window 40″ AFF.(Refer to section)”; side datum “Ground Floor Lvl - Garage.”	Measure bottom of actual emergency-opening clear opening from proposed finished floor, not old garage slab or an undefined frame sill. Reconcile raised floor on A01.6. Add W01 tags and certified opening data (7B).
5C Rewrite	Captions L{01}04 and L{01}05: “ELEVATION(Existing-New Window positions).”	Use “PROPOSED [verified façade] ELEVATION - EXISTING EXTERIOR WITH NEW W01 WINDOWS.” Label L{01}06 separately as existing/unchanged if intended. This removes uncertainty about proposed scope.
5D Add detail	Exterior faces around new windows; no keyed exhaust terminals/condensers shown in these elevations.	Show selected exhaust/utility penetrations, weatherproofing, retained finishes and new materials. Coordinate mechanical routes and outdoor equipment with site/plan (2D,4D). Name actual façades rather than “bottom side.”
5E Editorial	Right title summary “Ceiling Height 10′5″”; view scales 1/4″=1′; title-block 1:50; vertical “Roof Pivot lvl.”	Use verified height per 6A; reconcile scale per 3E; name the actual roof bearing/top-of-wall/ceiling datum after confirming intent. Do not call a roof-reference elevation the finished ceiling.

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Architect response / homeowner decision:_____

Document review only, not construction authorization or engineering/licensing approval. Original Review and source drawings unchanged.

A01.6 • Section

Source: “3571 Cassie Drive SW Submittal 13102026.pdf”, page 6. Review 2 page 6 corresponds to the same source page.

The section is especially important for coordinating the new floor, emergency openings, ramp and usable ceiling height.

ID / status	Exact point on the source page / current wording	Finding and requested correction / decision
6A Resolve	Right-hand vertical chain immediately left of grid 3 in L{01}02:1′4″+3′4″+4′+1′9″=10′5″; right title summary “Ceiling Height 10′5″.”	10′5″ starts at existing slab, not proposed floor. New floor to drawn roof datum is 9′1″ before ceiling/finish clarification. Replace summary: “Existing slab-to-roof datum 10′5″ as drawn; proposed finished ceiling height: verify and dimension after assemblies are coordinated.” Do not certify 9′1″ as finished ceiling.
6B Add assembly	Lower floor callout: “OSB boards on 2″ x8″ timber platform”; “Existing garage floor”; “New Floor Lvl.”	Show complete 16″ elevation buildup: supports/joists, species/grade, spacing/spans/anchorage, rated sheathing/finish, moisture/insulation and cavity strategy as applicable. Dimension connections to main-house doors/landings. Do not invent thickness/spacing to fill the gap.
6C Resolve	Bottom horizontal section dimension, grids 2–3: 21′1″.	Existing/proposed plan control is 20′5″:8″ conflict. Revise from verified canonical controls together with elevation 5A and plan 4A.
6D Define	Left-hand wall leaders “Skin Wall” and “Garage door.”	Specify actual insulated framed backing/infill assembly, structural support and air/moisture continuity at slab, jambs and head. Homeowner: confirm garage-door appearance retention. If retained nonoperable, document safe treatment of door/mechanisms and obtain design/city acceptance.
6E Editorial	Top “Existing roof / Pitch”; side “Roof Pivot lvl”; caption “Scale 5/8″=1′-0″”; title-block 1:20.	Identify actual roof/ceiling datum; give roof pitch if needed or mark verified existing unchanged. 5/8″=1′ corresponds to 1:19.2, not 1:20. Verify actual plotted scale; use consistent view/detail references.

Reply by ID: Item_____ □Agree □Clarify □Alternative Revised sheet/rev_____ Date_____
Architect response / homeowner decision:_____

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A01.7 • Window & Wall Details

Source: “3571 Cassie Drive SW Submittal 13102026.pdf”, page 7. Review 2 page 7 corresponds to the same source page.

The window schedule and its blank EGRESS column are the controlling points for product selection, not the reference photograph.

ID / status	Exact point on the source page / current wording	Finding and requested correction / decision
7A Resolve	Middle-right WINDOW SCHEDULE, W01 row: R/O61″×49″; HEADER “2X8X52(3)”. Upper-left NEW HEADER DETAIL says double 2×6/2×8/etc.	If 52 means inches of cut length, it is 9″ shorter than 61″ opening before bearing. Clarify notation/(3); schedule verified plies, section, length, grade/species, bearing/jack studs and connections/load basis. Generic reference detail cannot substitute for actual header design.
7B Verify	Same W01 row: SIZE 5′×4′; ACTUAL AREA 20 SF; EGRESS cell blank; upper-right WINDOW NOTES:44″,5.7 SF,24″,20″.	20 SF is nominal unit area, not slider clear opening. Rename column; fill manufacturer/model, net clear width/height/area and bottom-of-clear-opening AFF.20″×24″ is only 3.33 SF. Keep 5.7 SF unless grade-floor 5.0-SF exception is demonstrated; max opening height 44″ AFF.
7C Editorial	Upper-right WINDOW NOTES, last line: “HRC R310.1-R310.1.4.” Upper-left reference drawing labels “Jack Studs(Cripple Studs)”.	Use adopted IRC R310.2.1–.3 plus WA R330.6 for these opening criteria. Identify header-bearing jack/trimmer studs beside opening separately from cripple studs above header/below sill.
7D Add detail	Above schedule “VERTICAL FENESTRATION U=0.30 OR LOWER”; upper-right manufacturer-installation note; window photo.	U ≤ 0.30 is not an identified error. Add selected whole-unit rating, manufacturer-confirmed rough opening and head/jamb/sill weatherproofing integrated with existing WRB/cladding. Check safety glazing where applicable. Photo is illustrative, not a selected product specification.
7E Narrow scope	Bottom-left boxed note: “ALL NEW INTERIOR PARTITION WALL OR INFILL WALLING...”2×4@16 O.C.,1/2″ GWB each side; views L[01]07/08.	Limit generic wall note to dry nonbearing interior partitions. Separately detail exterior backing/infill, wet shower substrate/waterproofing, load/shear status and wall anchorage. Coordinate grab-bar backing.
7F Complete	Bottom-middle/right DOOR HARDWARE paragraph: one hand/no grasping, pinching, twisting.	Add 5-lb maximum activation force; bedroom/bath exterior release; closet interior release; graspable pocket-door hardware if used; required exit without added locks and key/tool-free reentry after exiting. Door widths are a separate schedule item (4C, page 10).

Reply by ID: Item_____ □Agree □Clarify □Alternative Revised sheet/rev_____ Date_____

Architect response / homeowner decision:_____

Document review only, not construction authorization or engineering/licensing approval. Original Review and source drawings unchanged.

A01.8 • Ramp Details

Source: “3571 Cassie Drive SW Submittal 13102026.pdf”, page 8. Review 2 page 8 corresponds to the same source page.

“36-inch landing” is not automatically wrong. This review does not assume ADA applies or impose a universal five-foot landing.

ID / status	Exact point on the source page / current wording	Finding and requested correction / decision
8A Resolve	Upper-left RAMP FRAMING detail 1: “RAMP TO RISE BY 6″”; cover/site 16″ and section 1′4″.	Verify final finished rise; show endpoints and horizontal sloped run separate from landings. At nominal 1:12:6″ needs 72″ run; 16″ needs 192″. These are arithmetic examples, not final slope approval; see 8B.
8B Coordinate	Under RAMP SECTION detail 2: “RAMP SLOPE 1:12”; boxed concrete-alternative note.	WA IRC maximum 1:12=8.3333%; applicable DSHS rule says ≤ 8.3% travel slope. Design to both with tolerance. At 8.3%,6″ needs ≥ 72.29″ and 16″ ≥ 192.77″ run. Prefer flatter practical geometry, not rounded equality.
8C Rewrite	Four small landing boxes around details 1/2: “LANDING SHOULD NOT BE LESS THAN 3 FT ON BOTH SIDES.”	Replace “should/both sides” with top, bottom, doors and direction-change landings; ≥ 36″ travel dimension, width ≥ ramp width. Show actual dimensions and door operation.3′ is not automatically deficient; final geometry must fit the layout.
8D Correct	Bottom-left CROSS SLOPE DETAIL 3: two notes “1:48 OR 2%” (ramp and landing).	1:48=2.0833%, not 2.0%. If intended limit is 2%, write “Maximum 2.0%(1:50).” Explicitly limit landing slope in travel direction to ≤ 2.0%; state chosen transverse limit clearly.
8E Add structure	Upper-left member labels 2×8 PT joist,4×6 support,4×4 post; middle-left “8×12 CONCRETE FOOTER(6 Total)”.	Supply spans/spacing/grade, decking, member/rail connections, post anchorage and soil/frost basis. Dimension complete footing width/length/thickness or diameter/depth. “Footer”→“footing.” Capacity not proven; not a finding that these sizes necessarily fail. Concrete option needs a complete alternate detail or deletion.
8F Rewrite	Bottom-right detail 4 boxed sentence: “...NOT BE LESS THAN 34″ AND MORE THAN 38″ (IRC 311.8.3)”; middle-right rail reference.	Use “Top of gripping surface 34″ minimum,38″ maximum vertically above finished ramp; IRC R311.8.3.1, both sides per R330.9.1.” Detail continuity/returns/anchors/clear width. Existing 12″ extensions may remain a design choice, not an assumed ADA/IRC mandate.
8G Editorial	RAMP SECTION: “2×2 BALUSTRADES SPACED AT 4″”; two details numbered 1; right reference says 36″ min clear/4′ overall.	Say balusters and identify clear gap versus centers; dimension selected guard/edge system and applicable drop-height requirements. Uniquely number details. Tie clear width to actual project framing, not only reference artwork.

Reply by ID: Item_____ □Agree □Clarify □Alternative Revised sheet/rev_____ Date_____

Architect response / homeowner decision:_____

Document review only, not construction authorization or engineering/licensing approval. Original Review and source drawings unchanged.

A01.9 • Grab Bar Details

Source: “3571 Cassie Drive SW Submittal 13102026.pdf”, page 9. Review 2 page 9 corresponds to the same source page.

Main AFH bar dimensions are generally sound; the missing items are project selection, support and coordinated installation.

ID / status	Exact point on the source page / current wording	Finding and requested correction / decision
9A Add backing	Top-right toilet perspective/section in detail 1: “Adequate Backing(Not Pictured)” / “Adequate Backing for Anchor Plates”.	Provide actual blocking/support, attachment to framing, fastener/embedment, product rating and finish/waterproof penetration allowance. R330 requires 250 lb at any point on bar, fastener, mounting device or supporting member. No direct support by residential-grade fiberglass shower/tub unit.
9B Checked	Upper-left/detail 1 and middle/detail 2 toilet drawings:25–30″ spacing; 33–36″ height; 28″ swing-up; 36″ fixed/12″ rear-wall offset.	Preserve these AFH values. Select actual fixed/swing-up combination and locate both-side WC bars on bathroom plan 4B. Dimension centerline spacing and finished-floor height for that selection, not just reference illustrations.
9C Checked	Lower-left SHOWER GRAB BAR DETAILS 3:30″×48″ minimum; 18″ vertical; bottom 36–42″; horizontals 33–36″; 4″ entry/6″ end offsets.	Main dimensions generally align with R330. Issue an actual selected shower plan/elevation showing finished clear interior, bars, controls, threshold, drain and waterproofing. Reference dimensions alone do not establish built enclosure clearances.
9D Select	Lower-right “MORE SHOWER DETAILS” control/seat/ICC-ANSI reference artwork.	Label/delete nonapplicable reference. If using permanent built-in-seat exception, add required floor-to-ceiling bar within 4″ of shower exterior aligned with seat nose. Otherwise show normal AFH vertical end bars and horizontals on all sides. Do not mix unrelated accessibility layouts.
9E Add spec	Bar diagrams throughout; product/support specification not supplied.	Specify circular grab-bar diameter 1¼–2″; fixed-bar wall spacing 1½″; clearance to controls/projections; graspable nonrotating surface. Distinguish swing-up spacing exception. All installation heights must use final finished floor.
9F Editorial	View captions “PLANS” under toilet/shower; “MORE SHOWER DETAILS”; general footer note 1.4 unfinished levels.	Use PLAN; title the selected shower configuration; add product tags/quantities, backing callouts and plan-to-detail keys. Override unfinished-level boilerplate for grab-bar/opening/handrail dimensions on every sheet.

Reply by ID: Item_____ □Agree □Clarify □Alternative Revised sheet/rev_____ Date_____
Architect response / homeowner decision:_____

Document review only, not construction authorization or engineering/licensing approval. Original Review and source drawings unchanged.

Coordination, responsibilities and reference key

One reply register for architect / homeowner. Read with the original, unchanged 18-page Review for expanded findings and bibliography.

Order: homeowner confirms use/occupancy, dryer and garage-door appearance; architect verifies site/levels and legal data, resolves dimensions, selects compliant products and completes assemblies/systems. Specialist engineering and city/DSHS acceptance remain required where applicable.

ID / status	Exact point on the source page / current wording	Finding and requested correction / decision
10A Shared footer/title block on every source sheet	Footer note 1.4: “...unfinished levels”; title summary 10′5″; CONSTRUCTION/Scheme Design/preliminary; repeated/mixed detail refs.	Use final finished surfaces for accessibility/opening dimensions. Correct height per 6A; one issue status; full owner/preparer/date/revision; verify licensing/seal obligations with city. Correct “discrepencies”; standardize AFF, PT, O.C., SF, NTS, WC/LAV, footing, baluster. Scales:1/4″=1′ is 1:48; 5/8″=1′ is 1:19.2. Verify printed views, not just rename them.
10B Missing whole-home evidence (cover 1B/plan 4C/site 2C)	Garage-only plan does not show all resident rooms, floors, facilities or route.	Add all-floor AFH plan with resident/household counts, usable room areas, S/NS1/NS2, designated exits/thresholds, stair-free route for assisted evacuation, safe location and whole-home alarms. Demonstrate five-minute evacuation; visual alarms where residents cannot hear warnings. Confirm toilet ratio 1 per 5 residents+household members.
10C Door/alarm code nuance (4C/4E/7F)	“3′ door” and “WAC R314.4” are not complete compliance specifications.	Required IRC egress: ≥ 32″ clear(face-to-stop at 90°), ≥ 78″ clear height. Current WAC 388-76-10715(6):internal resident-pass-through doors not emergency exits ≥ 27″ wide for homes licensed after 9/20/2026; not a universal 32″ bedroom rule. Use R330.4 hardware. Alarm basis:WA R314/R315 and R330.5; specify power/backup without ignoring applicable alteration exceptions.

Communication rule: reply “4A2 corrected on A01.4, rev___” rather than “dimensions fixed.” Do not assert a 5×4 slider fails, a footing/header is undersized, a three-foot landing fails, or ADA applies without the missing evidence.

Primary reference key (full sources in original Review)

Code/adoption
Tumwater adopted 2021 codes and WA amendments
https://www.ci.tumwater.wa.us/departments/community-development-department/permitting-building/building-codes

AFH building geometry/hardware
WAC 51-51-0330(IRC R330); egress/window/header:2021 IRC R310, R311.2, R311.8, R602.7
https://app.leg.wa.gov/wac/default.aspx?cite=51-51-0330

Bedroom usable area
WAC 388-76-10690:80 SF single/120 SF double, exclusions
https://app.leg.wa.gov/wac/default.aspx?cite=388-76-10690

Doors(current 2026 rule)
WAC 388-76-10715
https://app.leg.wa.gov/wac/default.aspx?cite=388-76-10715

Evacuation/ramp/visual alarms
WAC 388-76-10865:8.3% ramp/2% landing travel slopes
https://app.leg.wa.gov/wac/default.aspx?cite=388-76-10865

Energy conversion
WAC 51-11R-50200; fenestration table 51-11R-40211
https://app.leg.wa.gov/wac/default.aspx?cite=51-11R-50200

Smoke/CO
WAC 51-51-0314 and 51-51-0315
https://app.leg.wa.gov/wac/default.aspx?cite=51-51-0314

Tumwater’s linked AFH form still cites 2018 IRC; use current WA rules and adopted codes. Code facts carried forward from the original review; no new agency approval implied. Other referenced rules: WAC 388-76-10780(toilets),51-51-0315(CO),51-11R-40211(U-factor).9 unique source sheets; standalone files match combined pages 2–9.

Document review only, not construction authorization or engineering/licensing approval. Original Review and source drawings unchanged.

TUMWATER AFH CONVERSION — COMBINED MASTER BOOKLET

Value-Engineered Material Takeoff ($27.7k), Physical Construction Specs & City Inspection Guide

Property: 3571 Cassie Drive SW, Tumwater, WA 98512 | Permit Center / Inspection Scheduling: (360) 754-4180

EXECUTIVE SUMMARY & INSPECTION COORDINATION:
This booklet integrates the value-engineered $27,711.09 material takeoff budget with complete physical construction specifications and municipal inspection scheduling requirements for converting the 738 SF attached garage into a licensed 3-bedroom AFH. All City of Tumwater inspections (Rough, Insulation, Firewall, Final & WABO) are scheduled directly through the **Community Development Permit Center at (360) 754-4180**.

SECTION 1: ITEMIZATION OF MASTER MATERIALS TAKEOFF (VALUE-ENGINEERED < $30k)

#	Construction Trade	Key Material Scope Summary	Units	Trade Cost	Primary Store / Supplier
1	**Site Prep & Containment**	6-mil poly, zipper door kits, Ram Board floor protection	3 items	$330.00	Home Depot (Tumwater)
2	**Raised Floor & Wall Framing**	2x12 PT sleepers (16" rise), 3/4" OSB deck, R-30 insulation, 2x4 studs (10'-5"), PT bottom plates, 2x8 headers	14 items	$3,711.60	Builders FirstSource
3	**Shell, Windows & Doors**	1/2" CDX sheathing, 3 vinyl egress windows (5.7 SF clear), 1 utility window, 1 36" ADA exit door, flashing	5 items	$2,515.00	Builders FirstSource
4	**MEP Trade Rough-In**	PEX-A 1/2" & 3/4" lines, DWV PVC pipe 2" & 3", ASSE 1070 mixing valve, washer box, 14/2 & 12/2 wire, breakers	18 items	$1,782.40	Home Depot (Tumwater)
5	**Insulation & Soundproofing**	R-21 wall batts, R-15 perimeter batts, R-38 ceiling batts, Rockwool Safe'n'Sound acoustic batts, fireblock foam	6 items	$3,849.00	BFS / Lowe's Lacey
6	**Drywall & Fire Separation**	1/2" drywall (90 sheets), 5/8" Type X firecode (26 sheets for garage separation per IRC R302.6), greenboard, mud	8 items	$2,242.00	Home Depot Pro Desk
7	**Subfloor Leveling & LVP**	Self-leveler compound, primer, 37 boxes 20-mil commercial LVP (740 SF w/ waste), 6-mil poly, ADA reducers	6 items	$2,283.00	Home Depot (Tumwater)
8	**Tile, Waterproofing & Bath**	Cement backer, Schluter-Kerdi shower membrane, curbless shower pan kit, DCOF tile, subway tile, toilet, sink, grab bars	15 items	$2,958.00	Home Depot / Lowe's
9	**Doors, Locksets & Millwork**	5 solid-core 36" doors, 20-min fire door, bi-fold closet doors, ADA lever locksets, single-action exit lockset, MDF trim	12 items	$3,297.80	BFS / Home Depot
10	**HVAC, Vent & Heat Pump**	Quiet 80 CFM exhaust fan, R-6 flexible duct, 24k BTU multi-zone mini-split heat pump, branch ducting, registers	9 items	$2,504.00	Home Depot (Tumwater)
11	**Electrical Trim & Life Safety**	Tamper-resistant outlets, GFCI outlets, Decora switches, 6" LED wafer downlights, wet shower light, smoke/CO alarms	10 items	$978.80	Home Depot / Lowe's
12	**Paint, Finishes & Accessories**	PVA drywall primer, washable eggshell paint, semi-gloss trim enamel, painting supplies, closet shelves, bath accessories	6 items	$1,145.00	Home Depot (Tumwater)
13	**GC Verification Scopes**	Main electrical panel load calculation & HVAC Manual J equipment verification	2 HOLDs	HOLD	Home Depot / GC Audit
—	**BASE MATERIAL SUBTOTAL**	**All 110 priced material items above**	**110 Items**	**$24,096.60**	Base Materials Subtotal
—	**15% CONTINGENCY RESERVE**	**Material escalation & cutting waste buffer**	**15% Buffer**	**$3,614.49**	Contingency Buffer
—	**TOTAL MATERIAL BUDGET CEILING**	**Target Ceiling <= $30,000.00 Total**	**Target Met**	**$27,711.09**	**CEILING ACHIEVED**

SECTION 2: TRADE CONSTRUCTION SPECS & CITY INSPECTION PROTOCOLS

CITY OF TUMWATER BUILDING PERMIT & INSPECTION SCHEDULING DIRECTORY

Permit Center Phone: (360) 754-4180 | **Email:** cdd@ci.tumwater.wa.us | **Address:** 555 Israel Road SW, Tumwater, WA 98501
Inspection Request Lead Time: Call before 3:30 PM for next-business-day inspection. Permit # must be displayed on site.

Trade 1: 16" Raised Insulated Floor Platform & Wall Framing

1. Lay 6-mil poly sheeting over garage slab w/ 6" overlapped taped seams as a moisture break.
2. Anchor PT 2x12 floor sleepers @ 12" O.C. w/ 1/2" x 3-3/4" concrete wedge anchors to elevate deck 16 inches flush w/ main house.
3. Friction-fit R-30 subfloor insulation batts into sleeper joist cavities.
4. Apply subfloor adhesive to joist tops; glue & screw 3/4" T&G; OSB subfloor panels.
5. Frame interior 2x4 partition walls @ 16" O.C. (10'-5" ceiling height) w/ double top plates and PT bottom plates.
6. Frame 2x6 skin wall behind retained garage doors.
7. Install solid 2x8 horizontal wood blocking in bathroom wall cavities for commercial grab bar anchoring.

■ **CITY INSPECTION #1 (Underfloor Sleeper & Framing Check):** Call **(360) 754-4180** to verify sleeper anchors, PT plates, R-30 insulation & OSB deck prior to closing walls.

Trade 2: Exterior Shell Sheathing, Windows & ADA Egress Door

1. Fasten 1/2" CDX exterior plywood over garage door infill framing cavities.
2. Frame 2x8 structural headers for 3 egress window cutouts on side/rear walls and 1 HVAC utility window.
3. Apply self-adhering flashing tape around window rough openings.
4. Set vinyl slider egress windows; level & shim; verify sill height <= 44" AFF and clear net opening >= 5.7 sq ft per IRC R310.
5. Set 36" insulated exterior fiberglass door w/ low-profile ADA threshold; seal w/ polyurethane sealant.

Trade 3: MEP Rough-In (Plumbing, Electrical & HVAC)

1. Route 3" DWV sewer main to stack; run 2" branch drain lines for curbless shower and washing machine.
2. Pull 3/4" PEX-A main supply trunk & 1/2" branch runs; rough-in ASSE 1070 anti-scald thermostatic mixing valve (<=120°F).
3. Pull 14/2 NM-B wire for 15A bedroom circuits, 12/2 NM-B for 20A bath/laundry, and 14/3 for interconnected smoke alarms.
4. Set vapor-tight 1-gang outlet boxes and round ceiling junction boxes.

■ **CITY INSPECTION #2 (Rough MEP Inspection - Plumbing, Electrical, Mechanical):** Call **(360) 754-4180** before covering wall cavities. DWV pressure test (5 PSI) and electrical wire pull check required.

Trade 4: WSEC Thermal Insulation, Air-Sealing & Soundproofing

1. Seal all wall plate penetrations and outlet box edges w/ fireblock expanding foam.
2. Install R-21 Kraft wall batts in 2x6 infill wall, R-15 in perimeter walls, and R-38 in ceiling cavity (WSEC Zone 4C).
3. Install Rockwool Safe'n'Sound acoustic mineral wool batts in all resident bedroom partitions and bath walls.
4. Staple Class II smart vapor retarder membrane over exterior walls.

■ **CITY INSPECTION #3 (Insulation & Air Sealing Inspection):** Call **(360) 754-4180** to verify R-values, fireblocking & vapor barrier before sheetrock installation.

Trade 5: Drywall Boarding, Smooth Level 4 Taping & Fire Separation

1. Hang 5/8" Type X firecode drywall on garage-to-house separation wall & ceiling per IRC R302.6; tape w/ firecode compound.
2. Hang 1/2" greenboard moisture-resistant drywall in bathroom dry wall areas.
3. Hang 1/2" standard drywall across bedrooms, office, and hallway walls/ceilings.
4. Apply paper tape and 3 coats of joint compound to Level 4 smooth finish; apply acoustic sealant at floor perimeter seams.

■ **CITY INSPECTION #4 (Drywall Screw & Firewall Inspection):** Call **(360) 754-4180** to verify 5/8" Type X screw pattern and fire separation joint taping.

Trade 6: Waterproofing, Curbless Roll-In Shower & Tile Setting

1. Fasten 1/2" cement backer board to shower walls w/ alkali-resistant screws and mesh tape.
2. Install pre-sloped barrier-free curbless shower pan flush w/ OSB floor deck.
3. Apply Schluter-Kerdi waterproof sheet membrane up walls to 72" AFF; flood test for 24 hours.
4. Set DCOF >= 0.42 slip-resistant mosaic floor tile and white subway wall tile w/ polymer-modified thinset.
5. Apply polymer grout and 100% silicone sealant at wall/floor joint transitions.

Trade 7: Commercial 20-mil LVP & Millwork Trim Installation

1. Apply self-leveling compound at OSB floor seams; roll out 6-mil poly vapor underlayment film.
2. Install 20-mil commercial wear layer rigid core LVP w/ 1/4" wall expansion gaps; stagger end joints min 8".
3. Hang 36" solid molded core interior prehung doors (32" net clear passage); hang 20-min fire door at house entry.
4. Install Commercial Grade 2 ADA lever handles; install single-action interconnected exit lockset on exterior exit door.

5. Miter and nail primed MDF Craftsman baseboards (3-1/4") and casing (2-1/4").

Trade 8: MEP Trim Out, Life Safety Alarms & Final Finishes

1. Wire tamper-resistant duplex outlets, bath GFCI outlets, Decora rocker switches, and 6" LED wafer downlights.
2. Mount hardwired 120V interconnected smoke alarms inside Bedrooms A, B, C, and combo smoke/CO alarms in hallway/mech room.
3. Mount UL 924 battery-backup emergency exit light and 2 5-lb ABC fire extinguishers.
4. Apply 1 coat PVA drywall primer and 2 coats washable low-VOC eggshell paint on walls; enamel on trim.
5. Anchor ADA wall-hung lavatory sink, comfort-height toilet, 4 stainless grab bars into solid 2x8 wood backing, and phenolic folding shower seat.

■ **CITY INSPECTION #5 & WABO SIGN-OFF (Final Building, MEP & WABO Ready Inspection):** Call **(360) 754-4180** for final building, electrical, plumbing & mechanical signoff. Inspector signs DSHS Form 15-604 **WABO Checklist PASSED ('WABO Ready')** for AFH licensing.

SECTION 3: 8-WEEK MILESTONE, CASH FLOW & INSPECTION PAYMENT SCHEDULE

Draw #	Milestone Trigger & Inspection Focus	Material Order & Cost	GC Base Labor ($21.5k)	Specialty Trades ($ TBD)	10% GC Retainage	Net Payable Draw Amount
Draw 1	**Pre-Con Deposit & 16" Raised Floor**	Order 1: $3,850.00	$2,500.00	Included in GC	$250.00	**$6,100.00**
Draw 2	**Week 2: Shell & Egress Windows Set**	Order 2: $2,510.00	$3,000.00	Included in GC	$300.00	**$5,210.00**
Draw 3	**Week 3: Rough MEP Inspection (#360-754-4180)**	Order 3: $1,780.00	$3,500.00	Plumbing & Elec (Negotiating)	$350.00	**$4,930.00 + Sub**
Draw 4	**Week 4: Insulation Inspection (#360-754-4180)**	Order 4: $3,845.00	$2,000.00	Insulation Sub (Negotiating)	$200.00	**$5,645.00 + Sub**
Draw 5	**Week 5: Drywall Firewall Inspection (#360-754-4180)**	Order 5: $2,240.00	$3,500.00	Drywall Sub (Negotiating)	$350.00	**$5,390.00 + Sub**
Draw 6	**Week 6: Bath Tile & Shower Waterproof Test**	Order 6: $2,950.00	$3,000.00	Tile Sub (Negotiating)	$300.00	**$5,650.00 + Sub**
Draw 7	**Week 7: Doors, LVP & Trim Installed**	Order 7: $5,580.00	$2,500.00	Trim Sub (Negotiating)	$250.00	**$7,830.00 + Sub**
Draw 8	**Week 8: Final City Signoff & WABO Ready**	Order 8 + Reserve: $4,956.09	$1,500.00 (+$2,150 Release)	Final Sub Signoffs	-$2,150.00	**$8,606.09 + Sub***
TOTALS	**Turnkey Project Summary**	**$27,711.09 (Materials)**	**$21,500.00 (GC Base)**	**Under Negotiation**	**$2,150.00**	**$47,061.09 + Sub Labor**

**Note: Draw 8 includes full release of accumulated GC labor retainage ($2,150.00) and material contingency reserve upon final Certificate of Occupancy signoff and DSHS AFH WABO Ready certification. City of Tumwater Inspection Scheduling Phone: (360) 754-4180.*

TUMWATER AFH CONVERSION — COMBINED MASTER BOOKLET (V3)

Master Material Takeoff ($27.7k VE), Physical Construction Procedures & WABO Inspection Specs

Executive Summary & Core Project Specs:
Integrates the value-engineered $27,711.09 material takeoff budget w/ trade-by-trade physical construction procedures for converting the **790 SF attached 3-car garage** at 3571 Cassie Drive SW (Parcel #40380000700, Lot 7, Div 2) into a licensed 3-bedroom Adult Family Home.
• **16" Raised Insulated Floor Platform:** Eliminates ramps by elevating the 790 SF living space 100% level w/ primary house.
• **100% Elevation Preservation:** Retains garage doors w/ 2x6 skin walls behind; finished in matching lap siding (No T-111).
• **Financial Controls:** $21,500.00 GC base labor contract w/ 10% progressive retainage ($2,150.00 withheld) released upon WABO Ready signoff.

SECTION 1: TRADE-BY-TRADE CONSTRUCTION PROCEDURES & WABO SPECS

Trade 1 (Raised Floor Platform): Lay 6-mil poly over slab; anchor PT 2x12 sleepers @ 12" O.C. w/ 1/2" wedge anchors (16" rise); friction-fit R-30 underfloor batts; glue & screw 3/4" T&G; OSB deck; frame 2x4 partition walls @ 16" O.C. (10'-5" ceiling) and 2x6 skin wall.
Trade 2 (Shell & Openings): 1/2" CDX plywood; 3 egress windows (sill ≤ 44" AFF, clear opening ≥ 5.7 SF) + 1 utility window; 36" ADA exit door.
Trade 3 (MEP Rough): 3" DWV sewer & 2" drains; PEX-A supply; ASSE 1070 mixing valve (≤ 120°F); Romex wiring; vapor boxes. Call City line (360) 754-4180 for Rough MEP Inspection.
Trade 4 (Insulation): R-21 wall, R-15 perimeter, R-38 ceiling batts; Rockwool Safe'n'Sound acoustic batts; smart vapor retarder.
Trade 5 (Drywall): 5/8" Type X firecode drywall on garage separation wall/ceiling (IRC R302.6); Level 4 smooth finish.
Trade 6 (Waterproofing & Bath): Cement backer; curbless shower pan; Schluter-Kerdi membrane (24-hr flood test); DCOF ≥ 0.42 floor tile.
Trade 7 (LVP & Trim): Self-leveler; 20-mil commercial wear layer LVP (acclimated 48 hrs); 36" solid interior doors; Grade 2 ADA levers.
Trade 8 (Final Trim & WABO Signoff): Decora switches; 6" LED wafer downlights; hardwired smoke/CO alarms; UL 924 exit light; 2x 5-lb ABC fire extinguishers; wall-hung sink; comfort toilet; 4 grab bars. Call City for Final & **WABO Checklist Signoff ("WABO Ready")**.

SECTION 2: 8-WEEK CASH FLOW & WABO PAYMENT SCHEDULE

Draw #	Milestone Trigger	Material Scope & Cost	GC Base Labor ($21.5k)	Specialty Sub Status	10% Retainage	Net Payable Draw
Draw 1	Pre-Con & Raised Floor	Order 1: $3,850.00	$2,500.00	Included in GC	$250.00	**$6,100.00**
Draw 2	Week 2: Shell & Windows	Order 2: $2,510.00	$3,000.00	Included in GC	$300.00	**$5,210.00**
Draw 3	Week 3: Rough MEP Inspection	Order 3: $1,780.00	$3,500.00	Plumbing & Elec (Pending)	$350.00	**$4,930.00 + Sub**
Draw 4	Week 4: Insulation Inspection	Order 4: $3,845.00	$2,000.00	Insulation (Pending)	$200.00	**$5,645.00 + Sub**
Draw 5	Week 5: Drywall Level 4	Order 5: $2,240.00	$3,500.00	Drywall (Pending)	$350.00	**$5,390.00 + Sub**
Draw 6	Week 6: Bath Waterproofing	Order 6: $2,950.00	$3,000.00	Tile/Bath (Pending)	$300.00	**$5,650.00 + Sub**
Draw 7	Week 7: Doors, LVP & Trim	Order 7: $5,580.00	$2,500.00	Flooring/Trim (Pending)	$250.00	**$7,830.00 + Sub**
Draw 8	Week 8: **WABO Ready Signoff**	Order 8 + Contingency: $4,956.09	$1,500.00 (+$2.15k)	Final Sub Signoffs	-$2,150.00	**$8,606.09 + Sub***

CITY OF TUMWATER JOBSITE INSPECTION POSTING CARD

3571 Cassie Drive SW, Tumwater, WA 98512 | Parcel #40380000700 (Lot 7, Div 2)

MANDATORY POSTING NOTICE: Post this permit card in a weather-protected enclosure at the jobsite entrance alongside the City-stamped approved drawing set.

- **Permit Scope:** 790 SF garage-to-AFH conversion w/ 16" raised insulated floor platform & curbless shower bath.
- **Permit Valuation:** $52,711.09 | **Owners:** David Oh & Henry Oh | **Phone:** (564) 999-3464
- **City Scheduling Phone Line: (360) 754-4180** (Call before 3:30 PM for next-business-day inspection).

CITY INSPECTION SIGNOFF MATRIX

Inspection Stage	Target Week	Key Verification Scope	Inspector Signoff & Date
#101 Underfloor & Framing	Week 1	6-mil poly, PT sleepers @ 12" O.C., wedge anchors, R-30 batts, OSB deck.	[ ] PASS Date: _____ Inspector: _____
#201 Rough MEP Trades	Week 3	3" DWV pipe, PEX-A lines, ASSE 1070 mixing valve, NM-B wire, vapor boxes.	[ ] PASS Date: _____ Inspector: _____
#301 Insulation & Air Seal	Week 4	R-21 wall, R-15 perimeter, R-38 ceiling batts, Rockwool acoustic batts, foam.	[ ] PASS Date: _____ Inspector: _____
#401 Drywall & Firewall Check	Week 5	5/8" Type X firecode drywall screw pattern on separation wall/ceiling (R302.6).	[ ] PASS Date: _____ Inspector: _____
#501 Final & WABO Ready Signoff	Week 8	Final building finish, MEP trim, smoke/CO alarms, exit sign, WABO Ready signoff.	[ ] PASS Date: _____ Inspector: _____

JOBSITE MATERIAL DELIVERY RECEIPT & INTAKE LOG

3571 Cassie Drive SW, Tumwater, WA 98512 | Parcel #40380000700 (Lot 7, Div 2) | 790 SF Project

Material Intake & Storage Instructions:

• Inspect all incoming material drops for 790 SF garage-to-AFH conversion against supplier bills of lading.

• **Lumber & OSB:** Elevate off garage slab on 2x4 stickers; cover w/ waterproof tarps if staged outside.

• **Drywall & Insulation:** Store strictly indoors, 100% dry and elevated off slab.

• **20-mil LVP Flooring:** Acclimate inside covered conditioned space for **48 hours** prior to installation.

• **Supplier Contacts:** Home Depot Tumwater ((360) 704-7462), Lowe's Lacey ((360) 359-9044), Builders FirstSource Shelton ((360) 426-2611).